

School of Physical and Chemical Sciences

MAPH480

Conformal Geodesics in Asymptotically Flat Vacuum Spacetimes

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Abstract

We explore the conformal geometry of asymptotically flat vacuum spacetimes, derive the Generalised Conformal Field Equations (GCFE), the conformal geodesic equations, and discuss the significance of the conformal geodesic equations in defining the Conformal Gauss Gauge (CGG). Next, we plot a congruence of conformal geodesics with the initial data given in Friedrich (2003) for the Minkowski and Schwarzschild spacetimes onto their standard conformal diagrams, among other pictures, giving a visualisation of their behaviour that does not exist elsewhere. Different initial data is then explored on Schwarzschild; we showcase how the conformal geodesics, their level sets and the resultant CGG is effected by these choices, where data that is adapted to $i^{\pm}$ is highlighted. Then, we generalise the congruence in Friedrich (2003) to the Kerr spacetime, to try and construct a caustic free and smooth global covering.

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Declaration

I certify that the content of this report is my own work. All code used was written by me, based on previous code built by Lachlan Campion and Dr Chris Stevens. Unless otherwise stated, figures were numerically generated using this code. My supervisors, Dr Chris Stevens and Dr Mariem Magdy, helped by proofreading the report and offering feedback.

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1 Introduction

Einstein's theory of gravity is perhaps one of the most successful physical theories of all time. But even more than 100 years after its inception (Einstein, 1915) there are still new avenues to explore. In its most succinct form, it relates the matter content of the universe to a curvature in *spacetime* that describes how objects within it move. Let a spacetime be defined as a four-dimensional, smooth Lorentzian manifold, $(\mathcal{M}, g)$; that is a differentiable manifold, $\mathcal{M}$, equipped with a metric tensor, g that has signature (1,3) or (3,1). The curvature of that spacetime, described by the Riemann tensor, $R^a{}_{bcd}$, will then behave according to a set of ten partial differential equations, the *Einstein Field Equations* (EFE),

$$R_{ab} - \frac{1}{2}Rg_{ab} + \lambda g_{ab} = \kappa T_{ab}\,, \tag{1}$$

where R_{ab}, and R are the Ricci curvature tensor and Ricci scalar respectively, T_{ab} is the stress-energy tensor, $\kappa = 8\pi G/c^4$ is the Einstein gravitational constant and λ is the cosmological constant. These work together with the geodesic equations

$$u^a \nabla_a u^b = 0\,, \tag{2}$$

for a curve, x^a, parametrised by λ, with tangent vector, $u^a \equiv \frac{\mathrm{d}x^a}{\mathrm{d}\lambda}$, along with Levi-Civita connection, ∇. The solutions of (2), known as *geodesics*, describe the straight line paths that objects in the spacetime move along when acted upon only by gravity. Fundamentally, (1) and (2) form the basis of Einstein's theory of gravity.

As the EFE are coupled and highly non-linear, we must turn to analytic simplifications and numerical calculations in order to study spacetimes that approximate real world systems. One framework for studying isolated systems is based on a process known as conformal compactification, originally introduced for the analysis of gravitational wave sources. An overview of the current state of conformal techniques from the perspective of numerical relativity is given in Frauendiener (2000) but we will describe topics that are relevant to this report as necessary. Essentially, conformal geometry considers transformations that preserve angles but not necessarily distance, known as conformal transformations. Consider an equivalence class of metrics related to the given spacetime metric by a conformal transformation, known as the conformal class of the metric. The given spacetime and metric are called physical, as the given metric is typically taken to satisfy the EFE. With certain conditions on the physical spacetime, it is possible to use a member of the conformal class to define a compactified spacetime, which 'covers' the physical spacetime in finite distance. The compactified spacetime and metric are then known as unphysical, while the process of constructing an unphysical spacetime is called conformal compactification. We can reformulate the EFE for a physical spacetime into a set of equations on some unphysical spacetime, that are also invariant under conformal transformations, known as the generalised conformal field equations. These can be used to, for example, numerically consider global Cauchy problems on infinite spacetimes that would otherwise only be covered locally (Frauendiener, 2000). We can understand a large range of astronomical phenomena through the approximation of an isolated system, so these tools are a powerful and useful object of study.

This report is concerned with a generalisation of geodesics known as conformal geodesics that arise naturally in the study of conformal compactification. By looking at auto parallel curves with respect to a connection that is suited to conformal geometry, the Weyl

connection, we are given a class of curves invariant under conformal transformations. These are extremely important in analysis of the conformal structure of a spacetime. In particular, they are used in the construction of the conformal Gauss gauge, which fixes the gauge for the generalised conformal field equations by evolving data given on an initial hypersurface along conformal geodesics. We will centre investigations around the choice of initial data found in (Friedrich, 2003) that is shown to globally cover Schwarzschild smoothly, and has been used in numerical simulations of the Schwarzschild spacetime (Frauendiener & Stevens, 2021). By understanding how the conformal geodesics evolve for different initial data we learn more about the constructed conformal Gauss gauge.

1.1 Notation

Many different symbols will be used for tensor indices, summarised as follows:

- $(a, b, c, \dots)$ Latin indices denote abstract index notation (Penrose & Rindler, 1984).
- $(i, j, k, \dots)$ used for spatial abstract indices.
- $(\mu, \nu, \dots)$ Greek indices denote components in a coordinate basis.
- $(\alpha, \beta, \dots)$ used for spatial coordinate basis indices.
- $(A, B, \dots)$, and $(\mathcal{A}, \mathcal{B}, \dots)$ will be used to denote indices running from $0, 1$ and $2, 3$ respectively where necessary.

Bold face Latin indices will be used to stand for $(0, 1, 2, 3)$ or $(1, 2, 3)$ in the standard way such that in an arbitrary basis $\{e_0^a, e_1^a, e_2^a, e_3^a\}$ we have

$$T_{\boldsymbol{ab}} = T_{ab} e^a_{\boldsymbol{a}} e^b_{\boldsymbol{b}} \,.$$

For component indices Einstein summation convention is always applied unless stated otherwise. Going forward natural units ($c = 1 = G$) will be assumed. The metric signature adopted is $(+, -, -, -)$, and torsion, Riemann and Ricci tensors are defined to satisfy

$$\begin{aligned} 2\nabla_{[a}\nabla_{b]} f &= T_{ab}{}^{c} \nabla_c f \,, \\ \left(2\nabla_{[a}\nabla_{b]} - T_{ab}{}^{d}\nabla_d\right) Z^c &= R^c{}_{dab} Z^d \,, \\ R_{ab} &= R^d{}_{adb} \,, \end{aligned}$$

for arbitrary vector, Z^c, function, f, and affine connection, ∇. The Schouten tensor will be defined as

$$P_{ab} \coloneqq \frac{1}{2}\left(R_{(ab)} - \frac{1}{2}R_{[ab]} - \frac{1}{6} g_{ab} R\right) .$$

Connection coefficients are defined to satisfy

$$\nabla_{\boldsymbol{a}} e^a_{\boldsymbol{b}} = \Gamma_{\boldsymbol{ab}}{}^{\boldsymbol{c}} e^a_{\boldsymbol{c}} \,,$$

for some basis $e^a_{\boldsymbol{a}}$, where $\nabla_{\boldsymbol{a}} = e^a_{\boldsymbol{a}} \nabla_a$ in the abstract index notation. Where relevant, physical spacetime quantities with be referred to with a tilde, $\tilde{g}$, $\tilde{\mathcal{M}}$, while unphysical spacetime quantities will have no tilde, g, $\mathcal{M}$. In the necessary sections, we will denote quantities derived from a Weyl connection with a widehat, $\widehat{R}$. We use an underline, $\underline{h}$, to denote quantities constructed on a spacelike hypersurface, $\underline{S} \subset \tilde{\mathcal{M}}$.

2 Background

2.1 Gravitational waves

With the first direct observations of gravitational waves by LIGO being as recent as 2015 (Abbott et al., 2016), gravitational waves are of current interest but have a long history. A basic idealisation that is useful when studying a real world system emitting gravitational waves is the isolated system; we suppose that, where the cosmological constant is zero, the metric reduces to Minkowski spacetime at large distances from the source. Isolated systems were studied in the context of gravitational waves by numerous figures in the 50s and 60s including Bondi and Sachs in a string of papers (Bondi, 1960; Bondi et al., 1962a, 1962b; E. T. Newman & Unti, 1962; Sachs & Bondi, 1962). In these contexts, the radiative gravitational waveform is only well-defined at the infinite endpoint of null geodesics, and so in numerical calculations which operate on a finite region we must approximate quantities relating to the waveform, an overview of which is given in Bishop and Rezzolla (2016). Bondi and Sachs' group were able to show the asymptotic behaviour of these gravitational waves and that the source must lose energy by the emission of gravitational waves in the famous Bondi mass loss formula (Bondi et al., 1962a). Out of this group also came the discovery of the asymptotic symmetry group of coordinate transformations preserving the metric, known as the Bondi-Metzner-Sachs (BMS) group. It was of great interest that this group was strictly larger than the Poincaré group, the symmetry group for Minkowski space. In this way, the structure of these *asymptotically flat* spacetimes was starting to emerge as interesting in its own right.

2.2 Conformal spacetime

Throughout the developments on gravitational waves, E. Newman and Penrose (1962) were working on what would now be called the Newman-Penrose (NP) formalism, the spinor formulation of general relativity. Penrose was eventually led to the approach of studying Minkowski spacetime under an equivalence class of *conformal transformations* of the metric, where $g \to \Omega^2 g$ for some scalar field Ω (Penrose, 1964). In this regime, it is possible to treat the points at infinity (those of interest for asymptotic behaviour) as finite points. This essentially allows for analysis of asymptotic behaviours without limiting processes, which has advantages for both analytical and numerical approaches. The general treatment is developed in Frauendiener (2000) as follows: a *physical* spacetime under study has the infinite end points of null geodesics attached to the manifold to create an *unphysical* manifold. The unphysical metric is defined by a conformal rescaling of the physical metric such that the conformal factor, Ω, is vanishing on this boundary. Then the unphysical metric is well-defined at all points of the unphysical manifold, which is interiorly diffeomorphic to the physical manifold. This process is known as a *conformal compactification*.

To make this more precise, let us introduce some definitions. An *asymptotically simple* (Frauendiener, 2000) space time will be any smooth, time- and space-orientable spacetime, $(\tilde{\mathcal{M}}, \tilde{g})$ where there exists another smooth Lorentzian manifold $(\mathcal{M}, g)$ such that

1. $\tilde{\mathcal{M}}$ is an open submanifold of $\mathcal{M}$ with smooth boundary $\partial\tilde{\mathcal{M}} \equiv \mathscr{I}$.

2. There exists a smooth scalar field, Ω, on $\mathcal{M}$ so that $g = \Omega^2 \tilde{g}$ and $\Omega = 0$, $\mathrm{d}\Omega \neq 0$ on $\mathscr{I}$.

3. Every null geodesic in $\mathcal{M}$ has a future and past endpoint on $\mathscr{I}$. The collections of future and past endpoints on $\mathscr{I}$ allow us to define two distinct regions $\mathscr{I}^+$, and $\mathscr{I}^-$ for the future and past endpoints respectively. These collections will be called *future null infinity* and *past null infinity* respectively.

These two manifolds with their metrics are known as the physical and unphysical spacetimes respectively. The first two conditions allow us to think of the unphysical manifold as an extension of the first, that has a well defined metric, while the last condition is a kind of completeness condition to ensure all null geodesics reach infinity. Examples of asymptotically simple spacetimes include de Sitter and anti-de Sitter spacetimes, along with Minkowski spacetime itself. Generally we will presume that the conformal factor, Ω, is positive within the physical spacetime, this is an important condition that ensures the conformal transformation preserves the signature of the metric.

We will not be primarily concerned with asymptotically simple spacetimes, instead relaxing condition 3 somewhat. These *weakly asymptotically simple* spacetimes only require isometry to a asymptotically simple spacetime in the neighbourhood of $\mathscr{I}$. This allows us to include spacetimes like the Schwarzchild solution, which have some null-geodesics that don't take on endpoints in $\mathscr{I}$. An *asymptotically flat* spacetime is one that meets the requirements for a weakly asymptotically simple spacetime but also satisfies $\tilde{R}_{ab} = 0$ in a neighbourhood of $\mathscr{I}$. The unphysical spacetime admits an extension beyond the boundary of $\tilde{\mathcal{M}}$ in $\mathcal{M}$ which is not uniquely determined by the physical spacetime. However, we find that the boundary of $\tilde{\mathcal{M}}$ in $\mathcal{M}$ is uniquely determined by the conformal structure of $\tilde{\mathcal{M}}$, with causal character dependant on the cosmological constant (Frauendiener, 2000). We find that where the vacuum EFE hold near $\mathscr{I}$ and $\lambda = 0$, the Weyl tensor vanishes at $\mathscr{I}$ implying a total vanishing of curvature and thus we see a justification for calling these spacetimes asymptotically flat (Penrose & Bondi, 1965). We will find it useful to define the concepts of *spacelike infinity*, i^0, *future timelike infinity*, i^+ and *past timelike infinity*, i^-; these are the start/end points of spacelike and timelike geodesics respectively[1]. With this framework in place, it is easy to see that conformal compactification is a useful tool for analytic and numerical investigations of these asymptotically simple spacetimes; when studying properties best defined at infinity, conformal compactification allows us to use local differential geometry since the infinite boundary is represented as a regular submanifold on the unphysical spacetime. For numerical simulations, we avoid the problem of unphysical boundary conditions at infinity that can cause problems and we can resolve quantities defined at infinity on finite regions (Frauendiener, 2000).

2.3 Conformal Field Equations

In order to study gravitational systems in the unphysical spacetime, we need to reformulate the EFE for the physical spacetime into a system of equations involving quantities

[1]Note that we group the start/end points of spacelike infinity due to them being connected in Minkowski spacetime and similar spacetimes, and it is common practice to refer to timelike and spacelike infinity as points even where they manifest as hypersurfaces, in analogy with the natural compactification process on Minkowski spacetime (Frauendiener, 2000).

defined with respect to the unphysical metric. We will henceforth work on the assumption that the spacetime of interest is weakly asymptotically simple, vacuum spacetime[2]. This will clearly be an asymptotically flat spacetime. Then, the physical spacetime satisfies the vacuum Einstein Equations, (1) with $\tilde{T}_{ab} = 0$,

$$\tilde{R}_{ab} - \frac{1}{2}\tilde{R}\tilde{g}_{ab} + \lambda\tilde{g}_{ab} = 0\,, \tag{3}$$

or equivalently, using the trace-reversed field equations

$$\tilde{R}_{ab} = \lambda\tilde{g}_{ab}\,, \tag{4}$$

We would like to formulate (4) in terms of quantities in the unphysical spacetime and the conformal factor. A first attempt might involve directly computing the transformations and inserting (4). From that approach, observing (A.17), we get

$$R_{ab} = \frac{\lambda}{\Omega^2}g_{ab} - \frac{2}{\Omega}\nabla_a\nabla_b\Omega + g_{ab}\left(\frac{3}{\Omega^2}\nabla^c\Omega\nabla_c\Omega - \frac{1}{\Omega}\nabla^c\nabla_c\Omega\right)\,. \tag{5}$$

On the point set $\Omega = 0$ representing $\mathscr{I}$, (5) is singular, and this is a problem as $\mathscr{I}$ is where all the interesting asymptotic behaviour lies! We want to find a representation of the EFE that live in the physical spacetime, but are non-singular on $\mathscr{I}$. We are led to a working system by considering a different approach, described first by Friedrich (1981). For simplicity, we will follow the $\lambda = 0$ derivation which means that

$$\tilde{R}_{ab} = 0 = \tilde{R}\,, \tag{6}$$

$$\tilde{R}^a{}_{bcd} = \tilde{C}^a{}_{bcd}\,. \tag{7}$$

We will start by examining the second contracted Bianchi identity for the Weyl tensor, which takes the form as in (A.6) for the unphysical metric

$$\nabla_a C^a{}_{bcd} + 2\nabla_{[d}P_{c]b} = 0\,. \tag{8}$$

Since we have assumed a vacuum solution with zero cosmological constant for the physical metric, we find

$$\tilde{\nabla}_a\tilde{C}^a{}_{bcd} = \tilde{\nabla}_a C^a{}_{bcd} = 0\,. \tag{9}$$

This equation is the same as the zero rest-mass equation for a spin 2 particle but with the wrong conformal transformation properties. Let us define $K^a{}_{bcd} = \Omega^{-1}C^a{}_{bcd}$; then we can show that $K^a{}_{bcd}$ satisfies the same equation in the unphysical space

$$\nabla_a K^a{}_{bcd} = 0\,, \tag{10}$$

using the property (A.10). Then, (10) is invariant under a conformal mapping which is the behaviour we expect for the massless spin 2 particle equation. On $\mathscr{I}$, the vanishing of the conformal factor causes the definition of $K^a{}_{bcd}$ to fail; as the Weyl tensor also vanishes, there is a natural extension of the definition to the limit value at $\mathscr{I}$ which is

[2] Although it is indeed possible to derive the metric conformal field equations for non-vacuum asymptotically flat spacetimes, assuming strong enough fall off conditions for the matter content. See Friedrich (2014).

well-defined in general (Frauendiener, 2000). In this way, $K^a{}_{bcd}$ is useful for extracting information about the gravitational radiation at infinity. For this reason, $K^a{}_{bcd}$ is known as the *gravitational tensor*. With (10) as the central equation, we will try to formulate a system of equations that supplants the EFE using the unphysical metric system, that is also regular on $\Omega = 0$. If we decompose the Riemann tensor as in (A.11), then using the definition of the gravitational tensor we have

$$R^a{}_{bcd} = 2\left(g^a{}_{[c}P_{d]b} - \delta_{b[c}P_{c]}{}^a\right) + \Omega K^a{}_{bcd}\,, \tag{11}$$

while (10) tells us that

$$\nabla_a C^a{}_{bcd} = (\nabla_a \Omega K^a{}_{bcd}) = (\nabla_a \Omega)\, K^a{}_{bcd} + \Omega\,(\nabla_a K^a{}_{bcd}) = (\nabla_a \Omega)\, K^a{}_{bcd}\,,$$

which upon substitution into (8) yields

$$(\nabla_a \Omega)\, K^a{}_{bcd} = \nabla_c P_{db} - \nabla_d P_{cb}\,. \tag{12}$$

In using the unphysical equations we have an extra gauge freedom in the choice of conformal factor that will be incorporated into a gauge choice. We want to see how the conformal factor will evolve given the choice of initial data. Define the scalar s via

$$s = \frac{1}{4}\nabla^c \nabla_c \Omega + \frac{1}{24} R\Omega\,.$$

By inserting the relevant form of the EFE, (6), the transformation of the Ricci scalar, (A.18), can be written as

$$2\Omega s - \nabla_c \Omega \nabla^c \Omega = 0\,. \tag{13}$$

Starting with the transformation of the Ricci tensor, (A.17), and inserting the above, (13), then writing in terms of s we find

$$\nabla_a \nabla_b \Omega = s g_{ab} - \Omega P_{ab}\,, \tag{14}$$

which can be viewed as an equation for Ω. We can then find an integrability condition for (14) by the following process; take the covariant derivative on both sides, ∇_c, commute ∇_c with ∇_a with, and contract (using g_{ab}) on a, c. This process yields the equation

$$\nabla_a(\nabla_c \nabla^c \Omega) - R_{dc}{}^c{}_a \nabla^d \Omega = \nabla_a s - (\nabla^c \Omega) P_{ac} - (\nabla^c P_{ac})\Omega\,,$$

from which we apply the twice contracted second Bianchi identity, (A.7), to reduce to

$$\nabla_a s = -(\nabla^b \Omega) P_{ab}\,. \tag{15}$$

Equations (10), (11), (12), (14) and (15) together with constraint (13) constitute the *Metric Conformal Field Equations* (MCFE), here listed together as a system

$$\nabla_a K^a{}_{bcd} = 0\,, \tag{16a}$$
$$R^a{}_{bcd} = 2\left(g^a{}_{[c}P_{d]b} - \delta_{b[c}P_{c]}{}^a\right) + \Omega K^a{}_{bcd}\,, \tag{16b}$$
$$\nabla_c P_{db} - \nabla_d P_{cb} = (\nabla_a \Omega)\, K^a{}_{bcd}\,, \tag{16c}$$
$$\nabla_a \nabla_b \Omega = s g_{ab} - \Omega P_{ab}\,, \tag{16d}$$
$$\nabla_a s = -\nabla^b \Omega P_{ab}\,, \tag{16e}$$
$$2\Omega s = \nabla_c \Omega \nabla^c \Omega\,. \tag{16f}$$

These equations are regular for $\Omega = 0$ and conformally invariant. They work as a replacement for (6), formulated entirely in terms of unphysical quantities. There is a problem with these equations as they stand however; the conformal factor is not known until we solve the system, making it hard to find the conformal boundary in advance. It turns out there is a generalisation of these equations where it is possible to solve this problem, described in Section 2.5. To derive this generalisation, we need to introduce a different type of connection that is more suited to conformal geometry, namely, the Weyl connection.

2.4 Weyl Connection

We are led to define a *Weyl connection*, $\widehat{\nabla}$, as a generalisation of the Levi-Civita connection by dropping the metric compatibility. Instead, take $\widehat{\nabla}$ to be a torsion-free affine connection satisfying

$$\widehat{\nabla}_a g_{bc} = -2h_a g_{bc}\,, \tag{17}$$

for some one-form h_a. This represents the most general (torsion-free) connection that preserves angles under parallel transport but is not-necessarily Levi-Civita to any metric in the conformal class (Akivis & Goldberg, 2011). In this way the Weyl connection is more suited to conformal geometry, where we look at properties preserved under preservation of angles but not distance. Under a conformal mapping $g_{bc} \to \theta^2 g_{bc}$ we find

$$h_a \to h_a - \theta^{-1}\widehat{\nabla}_a \theta\,, \tag{18}$$

via (17). Given two Weyl connections

$$\widehat{\nabla}_a g_{bc} = -2h_a g_{bc}\,,$$
$$\mathring{\nabla}_a g_{bc} = -2b_a g_{bc}\,,$$

with one-forms h_a and b_a, we define the difference one-form, f_a, by

$$\widehat{\nabla}_a g_{bc} - \mathring{\nabla}_a g_{bc} = -2h_a g_{bc} + 2b_a g_{bc} = -2f_a g_{bc}\,.$$

so that $f_a = h_a - b_a$. When the Weyl connections $\widehat{\nabla}$ and $\mathring{\nabla}$ are taken to be the Levi-Civita connections for the unphysical and physical space respectively, we recover the transformation law (A.10) for the action on vectors otherwise we have the generalised transformation law replacing ${\Upsilon_{ab}}^c$ with

$${f_{ab}}^c \coloneqq \delta^c_a f_b + \delta^c_b f_a - g_{ab}g^{cd}f_d\,. \tag{19}$$

The curvature tensors of the Weyl connection are defined in the standard way but we note that the change of connection means that we do not have all the typical symmetries. In particular $\widehat{R}_{ab}$ is not necessarily equal to $\widehat{R}_{(ab)}$. The decomposition of the Riemann tensor into the Weyl tensor and Schouten tensor takes the same form, given that Schouten tensor is defined in general by

$$\widehat{P}_{ab} \coloneqq \frac{1}{2}\left(\widehat{R}_{(ab)} - \frac{1}{2}\widehat{R}_{[ab]} - \frac{1}{6}g_{ab}\widehat{R}\right)\,. \tag{20}$$

This also ensures that the transformation laws derived in Appendix A take the same form. Note that (20) reduces to the standard definition of the Schouten tensor for a Levi-Civita connection.

2.5 Generalised Conformal Field Equations

With the Weyl connection now understood we return our attention to the construction of a set of equations that formulate the EFE in the unphysical spacetime. Let $\tilde{g}$ be a solution to the vacuum EFE with $\lambda = 0$ and $g = \Theta^2\tilde{g}$ be an unphysical metric for some conformal factor Θ. We start by defining the one-forms f and b by the requirements

$$\widehat{\nabla}_a \tilde{g}_{bc} = -2b_a \tilde{g}_{bc}\,, \tag{21}$$

$$\widehat{\nabla}_a g_{bc} = -2f_a g_{bc}\,, \tag{22}$$

where it follows from (18) that

$$f_a = b_a - \Theta^{-1}\widehat{\nabla}_a\Theta\,.$$

It is convenient to introduce a basis $e^a_{\boldsymbol{a}}$, that is orthonormal with respect to the conformal metric $\eta_{\boldsymbol{ab}} = g_{ab}e^a_{\boldsymbol{a}}e^b_{\boldsymbol{b}}$, and coordinates x^μ. These can be related through the frame coefficients

$$e^\mu_{\boldsymbol{a}} \coloneqq \widehat{\nabla}_{\boldsymbol{a}} x^\mu = e^a_{\boldsymbol{a}}\widehat{\nabla}_a x^\mu\,,$$

which give the coefficients of the orthonormal basis expressed in the coordinate basis. By using the unphysical metric to contract (22) on b, c and expressing in terms of the orthonormal basis, we find

$$f_{\boldsymbol{a}} = \frac{1}{4}\widehat{\Gamma}_{\boldsymbol{ab}}{}^{\boldsymbol{b}}\,.$$

Next, we seek to express the torsion-free conditions and curvature equations in terms of our basis. The torsion-free condition is given by

$$2\widehat{\nabla}_{[a}\widehat{\nabla}_{b]}x^\mu = 0\,,$$

which, in components, gives the equation

$$\widehat{\nabla}_{\boldsymbol{a}} e^\mu_{\boldsymbol{b}} - \widehat{\nabla}_{\boldsymbol{b}} e^\mu_{\boldsymbol{a}} = \widehat{\Gamma}_{\boldsymbol{ab}}{}^{\boldsymbol{c}} e^\mu_{\boldsymbol{c}} - \widehat{\Gamma}_{\boldsymbol{ba}}{}^{\boldsymbol{c}} e^\mu_{\boldsymbol{c}}\,. \tag{23}$$

The curvature equation acting on the basis vectors gives the equation

$$\left(\widehat{\nabla}_a\widehat{\nabla}_b - \widehat{\nabla}_b\widehat{\nabla}_a\right) e^c_{\boldsymbol{d}} = \widehat{R}^c{}_{dab} e^d_{\boldsymbol{d}} = \widehat{R}^c{}_{\boldsymbol{d}ab}\,,$$

which, in components, after utilising the (A.1) gives the equation,

$$\widehat{\nabla}_{\boldsymbol{a}}\widehat{\Gamma}_{\boldsymbol{bc}}{}^{\boldsymbol{d}} - \widehat{\nabla}_{\boldsymbol{b}}\widehat{\Gamma}_{\boldsymbol{ac}}{}^{\boldsymbol{d}} = 2\widehat{\Gamma}_{\boldsymbol{[ab]}}{}^{\boldsymbol{e}}\widehat{\Gamma}_{\boldsymbol{ec}}{}^{\boldsymbol{d}}\widehat{\Gamma}_{\boldsymbol{[a|c|}}{}^{\boldsymbol{e}}\widehat{\Gamma}_{\boldsymbol{b]e}}{}^{\boldsymbol{d}} - \Theta K_{\boldsymbol{abc}}{}^{\boldsymbol{d}} - 2\eta_{\boldsymbol{c[a}}\widehat{P}_{\boldsymbol{b]}}{}^{\boldsymbol{d}} + 2\delta^{\boldsymbol{d}}_{\boldsymbol{[a}}\widehat{P}_{\boldsymbol{b]c}}\,. \tag{24}$$

Finally, we investigate the second Bianchi identity for the Weyl connection

$$\widehat{\nabla}_{[e}\widehat{R}_{ab]c}{}^d = 0\,,$$

which can be contracted on d, e to give

$$\widehat{\nabla}_e\widehat{R}_{abc}{}^e + \widehat{\nabla}_a\widehat{R}_{bc} - \widehat{\nabla}_b\widehat{R}_{ac} = 0\,, \tag{25}$$

where here, and in the following derivations, indices are raised/lowered with unphysical metric g_{ab}. Decomposing the Riemann tensor, (A.12), and writing the Ricci tensor in terms of the Schouten tensor via (A.13) we find from (25)

$$\widehat{\nabla}_e C_{abc}{}^e + 2g_{c[a}\left(\widehat{\nabla}^e \widehat{P}_{b]e} - \widehat{\nabla}_{b]}\widehat{P} + 2f_{b]}\widehat{P}\right) + 2\widehat{\nabla}_{[a}\widehat{P}_{b]c} = 0\,. \tag{26}$$

Upon metric contraction of (26) with g on b, c, we find

$$4\left(\widehat{\nabla}^e \widehat{P}_{ae} - \widehat{\nabla}_a \widehat{P} + 2f_a \widehat{P}\right) = 4\widehat{\nabla}^e\left(\widehat{P}_{ae} - g_{ae}\widehat{P}\right) = 0\,, \tag{27}$$

and inserting (27) into (26) wipes out the centre term to give

$$\widehat{\nabla}_e C_{abc}{}^e + 2\widehat{\nabla}_{[a}\widehat{P}_{b]c} = 0\,. \tag{28}$$

To complete the the above equation, we can use (A.10) with $f_{ab}{}^c$ to find (10) in terms of a Weyl connection

$$\widehat{\nabla}_e K_{abc}{}^e = f_e K_{abc}{}^e\,,$$

and finally write

$$\left(\widehat{\nabla}_e \Theta + \Theta f_e\right) K_{abc}{}^e + 2\widehat{\nabla}_{[a}\widehat{P}_{b]c} = 0\,. \tag{29}$$

Equations (23), (24), (29), along with (10), together constitute the *Generalised Conformal Field Equations* (GCFE), here listed together as a system

$$\widehat{\nabla}_{\boldsymbol{a}} e^\mu_{\boldsymbol{b}} - \widehat{\nabla}_{\boldsymbol{b}} e^\mu_{\boldsymbol{a}} = \widehat{\Gamma}_{\boldsymbol{ab}}{}^{\boldsymbol{c}} e^\mu_{\boldsymbol{c}} - \widehat{\Gamma}_{\boldsymbol{ba}}{}^{\boldsymbol{c}} e^\mu_{\boldsymbol{c}}\,, \tag{30a}$$

$$\begin{aligned}\widehat{\nabla}_{\boldsymbol{a}}\widehat{\Gamma}_{\boldsymbol{bc}}{}^{\boldsymbol{d}} - \widehat{\nabla}_{\boldsymbol{b}}\widehat{\Gamma}_{\boldsymbol{ac}}{}^{\boldsymbol{d}} &= 2\widehat{\Gamma}_{\boldsymbol{[ab]}}{}^{\boldsymbol{e}}\widehat{\Gamma}_{\boldsymbol{ec}}{}^{\boldsymbol{d}}\widehat{\Gamma}_{\boldsymbol{[a|c|}}{}^{\boldsymbol{e}}\widehat{\Gamma}_{\boldsymbol{b]e}}{}^{\boldsymbol{d}} - \Theta K_{\boldsymbol{abc}}{}^{\boldsymbol{d}} \\ &\quad - 2\eta_{\boldsymbol{c[a}}\widehat{P}_{\boldsymbol{b]}}{}^{\boldsymbol{d}} + 2\delta^{\boldsymbol{d}}_{\boldsymbol{[a}}\widehat{P}_{\boldsymbol{b]c}}\,,\end{aligned} \tag{30b}$$

$$\left(\widehat{\nabla}_e \Theta + \Theta f_e\right) K^e{}_{abc} - 2\widehat{\nabla}_{[a}\widehat{P}_{b]c} = 0\,, \tag{30c}$$

$$\widehat{\nabla}_a K^a{}_{bcd} - f_a K^a{}_{bcd} = 0\,. \tag{30d}$$

These equations are regular for $\Theta = 0$ and invariant under change of Weyl connection or conformally transformed metric. They work as a replacement for (6), formulated entirely in terms of unphysical quantities. Given a suitable gauge, it turns out they also give the conformal factor in advance. In order to define this gauge, we introduce the key conformal structure in this report, conformal geodesics.

2.6 Conformal Geodesics

We are motivated to introduce a generalisation of geodesics via the Weyl connection, and these are known as *conformal geodesics*. Instead of using auto parallel curves with respect to a Levi-Civita connection, we want to have a system of equations that gives a curve that is auto parallel with respect to some Weyl connection

$$u^b \widehat{\nabla}_b u^a = 0\,. \tag{31}$$

We would also like this system to be invariant to changes in connection (in some sense) and invariant under conformal transformations. By relating (31) to a Levi-Civita connection via (A.10) we can write

$$u^b \nabla_b u^a + 2\left(b_c u^c\right) u^a - \left(u_c u^c\right) b^a = 0\,, \tag{32}$$

for some b_a such that $\widehat{\nabla}_a g_{bc} = -2b_a g_{bc}$. Under a change of connection $\nabla \to \mathring{\nabla}$, again via (A.10), we find

$$u^b\nabla_b u^a + 2\left(b_c u^c\right)u^a - \left(u_c u^c\right)b^a = u^b\mathring{\nabla}_b u^a + 2\left[\left(b_c - h_c\right)u^c\right]u^a - \left(u_c u^c\right)\left(b^a - h^a\right) ,$$

where h_a is such that $\mathring{\nabla}_a g_{bc} = -2h_a g_{bc}$. That is, the expression (32) is invariant under change of connection where the one-form transforms as $b_a \to b_a - h_a$. Seeking an equation to pick a one-form and thus a Weyl connection, we take the condition

$$u^a \widehat{P}_{ab} = 0 . \tag{33}$$

Relating the Schouten tensor to the Levi-Civita connection via (A.14), we see that

$$u^b\nabla_b b_a - \left(b_c u^c\right)b_a + \frac{1}{2}b_c b^c u_a + P_{ab}u^b = 0 , \tag{34}$$

Then, under a change of connection $\nabla \to \mathring{\nabla}$, once again via (A.10), we find

$$\begin{aligned} &u^b\nabla_b b_a - \left(b_c u^c\right)b_a + \frac{1}{2}b_c b^c u_a + P_{ab}u^b \\ =&u^b\mathring{\nabla}_b\left(b_a - h_a\right) - \left[\left(b_c - h_c\right)u^c\right]\left(b_a - h_a\right) + \frac{1}{2}\left(b_c - h_c\right)\left(b^c - h^c\right)u_a + \mathring{P}_{ab}u^b , \end{aligned}$$

and (34) is invariant under the change of connection in the same sense that (32) was. Then we can define conformal geodesics as a tangent vector, u^a, describing a curve, γ, along with a one-form along γ, b_a, that satisfy the Conformal Geodesic Equations (CGE):

$$u^b\nabla_b u^a = -2\left(b_c u^c\right)u^a + \left(u_c u^c\right)b^a , \tag{35a}$$

$$u^b\nabla_b b_a = \left(b_c u^c\right)b_a - \frac{1}{2}\left(b_c b^c\right)u_a - P_{ab}u^b . \tag{35b}$$

The conformal geodesic equations are suited to the conformal structure as they are invariant under changes in connection and conformal transformations, in the sense that $(u^a, b_a - f_a)$ is a solution to the transformed equations where f_a is the difference one-form in the case of a connection change, or $f_a = \theta^{-1}\nabla_a\theta$ in the case of a conformal transformation. Since these equations admit a solution with $b_a = 0$, we find that metric geodesics form a subclass of conformal geodesics. Finally, in order to define how a frame is dragged along the curve we let a frame $e^a_{\boldsymbol{a}}$ be parallel transported by $\widehat{\nabla}$,

$$u^b\widehat{\nabla}_b e^a_{\boldsymbol{a}} = 0 , \tag{36}$$

to find

$$u^b\nabla_b e^a_{\boldsymbol{a}} = -\left(b_c u^c\right)e^a_{\boldsymbol{a}} - \left(b_c e^c_{\boldsymbol{a}}\right)u^a + \left(u_c e^c_{\boldsymbol{c}}\right)b^a . \tag{37}$$

On a physical spacetime, $\tilde{g}_{ab}$, that satisfies the vacuum EFE with $\lambda = 0$ (meaning $\tilde{P}_{ab} = 0$), the CGE, (35), imply the following for the evolution of various important scalar quantities along the geodesics

$$u^c\tilde{\nabla}_c\left(\tilde{g}_{ef}u^e u^f\right) = \left(\tilde{g}_{ef}u^e u^f\right)\left(b_c u^c\right) , \tag{38a}$$

$$u^c\tilde{\nabla}_c\left(b_a u^a\right) = -\left(b_c u^c\right)^2 + \frac{1}{2}\left(b_a b_c\tilde{g}^{ac}\right)\left(\tilde{g}_{ef}u^e u^f\right) , \tag{38b}$$

$$u^c\tilde{\nabla}_c\left(\tilde{g}^{ab}b_a b_b\right) = b_c u^c\left(\tilde{g}^{ab}b_a b_b\right) . \tag{38c}$$

On inspection of the CGE, they are essentially a third order system. It is possible to write this in an explicit way, and thereby extract the physical meaning of the one form as related to the covariant derivative of the tangent vector. Tod (2012) gives this *third order formulation* as follows: Start by defining unit tangent vector $v^a = \chi^{-2}u^a$, where $\chi^4 = g_{ef}u^e u^f > 0$, assuming it is time-like. We will only derive the time-like case for simplicity but the equations differ only by a sign for the space-like case. We write (35a) in terms of the acceleration $a^c = v^a\nabla_a v^c$ to find

$$b^c = a^c + \left(-2\chi^{-1}v^a\nabla_a\chi\right)v^c\,, \tag{39}$$

where we can find that $v^a b_a = -2\chi^{-1}v^a\nabla_a\chi$. Next, use this in (35a) to arrive at the third order formulation

$$v^c\nabla_c a^b = v^b\left(-g_{ef}a^e a^f - P_{ef}v^e v^f\right) + P^b_c v^c\,. \tag{40}$$

In order to constrain the normalisation factor, χ, we also find that

$$v^c\nabla_c\left(v^a b_a\right) = \frac{1}{2}\left(g_{ef}a^e a^f + \left(v^a b_a\right)^2\right) + P_{ef}v^e v^f\,,$$

which can be written succinctly as

$$v^c\nabla_c\left(v^a\nabla_a\chi\right) = -\frac{1}{4}\left(g_{ef}a^e a^f + 2P_{ef}v^e v^f\right)\chi\,. \tag{41}$$

By solving (40) for v^a, we can find a solution to the standard CGE, (u^a, b_a), by solving (41) for a choice of χ and using (39) to obtain b_a. The initial condition for b_a is essentially prescribing an acceleration for the coordinates along the tangent vector. In this way we see a more physical interpretation of the data given by the one-form.

2.7 Conformal Gauss Gauge

The GCFE as listed have invariance under a number of transformations, they are built from tensors and as such have invariance under coordinate and basis transformations, and are also invariant under conformal rescaling and changes of Weyl connection by construction. Each of these invariances give a gauge freedom to be fixed in order to use these equations in a real system. Of interest is the *Conformal Gauss Gauge* (CGG) which fixes the coordinates, basis, conformal factor, and one form given an initial space-like hyper surface. This gauge is particularly useful as it is obtained from the conformal structure, and happens to give the conformal factor in advance. It is built in a similar manner to the Gauss gauge for geodesics, where the frame and coordinates are chosen on an initial hypersurface, and evolved along the geodesics. Gauss gauges in curved spacetimes tend to develop overlapping points known as caustics; this is a problem as caustics stop the evolution of the frame and coordinates from being well-defined. Conformal geodesics have the additional freedom in the one-form parameter that can act to keep the conformal geodesics separated and avoid forming caustics globally. It makes sense that this might be possible, as we saw the one-form was related to an acceleration in Section 2.6.

Given a space-like hypersurface, $\underline{S}$, in the physical spacetime, $(\tilde{\mathcal{M}}, \tilde{g})$, choose conformal factor, $\underline{\Theta}$, one-form, $\underline{b}_a$, coordinates, $\underline{x}^\mu$, and a frame $\underline{e}^a_{\boldsymbol{a}}$, on this hypersurface (referred to as initial conditions). We require that the frame is orthonormal with respect to the conformal

metric defined by the conformal factor, $\underline{g}_{ab} = \underline{\Theta}^2 \tilde{\underline{g}}_{ab}$, and $\underline{e}^a_{\mathbf{0}}$ is chosen to be normal to $\underline{S}$. With the initial data $\{\underline{u}^a = \underline{e}^a_{\mathbf{0}}, \underline{b}_a\}$ we can find a solution, (u^a, b_a), to the conformal geodesic equations, (35), where the underlying metric is from the physical spacetime. Provided the congruence is smooth and caustic free in a neighbourhood $U \subset \mathcal{M}$ of $\underline{S}$ we can extend the coordinates and frame to U as follows: set x^0 to be the parameter of the conformal geodesics, τ, while dragging spatial coordinates along the conformal geodesics, set the frame to be transported along the geodesics according to the frame drag equation (37). Finally, we can extend the conformal factor to U by requiring that $g_{ab} e^a_{\boldsymbol{a}} e^b_{\boldsymbol{b}} = \eta_{\boldsymbol{ab}}$ in U. The extended coordinates, conformal factor, one-form, and frame define the CGG on U. We can write the gauge fixing conditions on x^μ and $e^a_{\mathbf{0}}$ as

$$u^a \tilde{\nabla}_a x^\alpha = 0\,, \qquad u^a \tilde{\nabla}_a \tau = 1\,, \qquad e^a_{\mathbf{0}} = u^a\,, \qquad \tilde{g}_{ab} u^a u^b = \Theta^{-2}\,. \tag{42}$$

We can proceed to calculate the derivatives of the conformal factor, given that the physical spacetime is a vacuum solution of the EFE with $\lambda = 0$. Use the above, (42), and (38a), to see

$$u^c \tilde{\nabla}_c \Theta \coloneqq \dot{\Theta} = \Theta \left(b_c u^c\right)\,. \tag{43}$$

Taking a derivative of (43) and using (38b) we find the second derivative, while (38c) allows us to compute the third derivative, and imputing (42) into these expressions gives

$$\ddot{\Theta} = \frac{1}{2\Theta} b_e b_f \tilde{g}^{ef}\,, \tag{44a}$$

$$\dddot{\Theta} = 0\,. \tag{44b}$$

This lets us write the conformal factor Θ as a quadratic function of the initial conditions

$$\Theta(\tau) = \underline{\Theta} + \underline{\dot{\Theta}}\tau + \frac{1}{2}\underline{\ddot{\Theta}}\tau^2 = \underline{\Theta} + \left(\underline{b}_c \underline{u}^c \underline{\Theta}\right)\tau + \left(\frac{1}{4\underline{\Theta}} \underline{b}_e \underline{b}_f \tilde{\underline{g}}^{ef}\right)\tau^2\,, \tag{45}$$

The result (45) is very useful, as it allows us to know the conformal boundary in advance. With the machinery now in place, the CGG can be used to set up numerical simulations of, for example, gravitational wave models that have all the advantages of conformal methods described in Section 2.2. A global covering of Schwarzschild for specific initial data on is shown to be regular and caustic free in Friedrich (2003), and this global cover has been used in numerical simulations such as Frauendiener and Stevens (2021). In Section 3 we first confirm and visualise the congruence resulting from this choice of initial data and investigate alternate choices with Minkowski and Schwarzschild as the physical spacetimes. Then, we look at generalising the initial data to the Kerr spacetime.

3 Numerical Implementation

Here, we numerically compute the conformal geodesics for various different initial data sets. It is known analytically that the Schwarzschild spacetime can be covered with a congruence of timelike conformal geodesics globally and without caustics, using initial data as given in Friedrich (2003), thus enabling the construction of the CGG globally. We numerically implement the conformal geodesic equations for this data on Minkowski and Schwarzschild. The CGG is built from the physical spacetime but when solving the CGE we can use any metric in the conformal class; we exploit this to evolve the conformal geodesics through $\mathscr{I}$. For Minkowski space, we can utilise the standard embedding into the Einstein cylinder. For Schwarzschild, Friedrich (2003) suggests some specific unphysical metric choices such that the solutions evolve smoothly through $\mathscr{I}^+$ and $\mathscr{I}^-$ separately. In both cases, the congruence is visualised on a conformal Penrose diagram.

Next, we explore different choices of initial data for the CGG. Assuming that the congruence is timelike, and evolves in spherical symmetry we are left the freedom to choose initial values for the one-form and conformal factor. It is of interest to find sets of initial data that reach $i^{\pm}$ or close to it before reaching the singularity or other points of $\mathscr{I}$. For example, the numerical setup modelling gravitational wave perturbations on Schwarzschild space, given in Frauendiener and Stevens (2021), requires evolution as close to i^+ as possible but the system breaks when we reach the singularity or venture too far into the unphysical region outside $\mathscr{I}$.

Finally, we adapt Friedrich's initial data for the Kerr spacetime in order to try and construct a caustic free congruence on the Kerr spacetime.

The visualisation can help with qualitative understanding of the behaviour of conformal geodesics, while new initial data sets can be used to improve existing numerical simulations of gravitational wave perturbations on Schwarzschild (Frauendiener & Stevens, 2021) and allow for extension the to Kerr spacetime.

In performing numerical tests and algebraic calculations, we used Mathematica and the Riemannian Geometry and Tensor Calculus Mathematica package ("Riemannian Geometry and Tensor Calculus @ Mathematica", 2013). In the following we take underlined quantities to be defined on an initial hypersurface $\underline{S}$.

3.1 Minkowski Space

We start with the Minkowski metric, which describes totally flat space. In polar coordinates, (t, r, ϕ, θ), the Minkowski metric is given by

$$\tilde{g} = \mathrm{d}t^2 - \mathrm{d}r^2 - r^2\mathrm{d}\sigma^2 \,,$$

where $\mathrm{d}\sigma^2$ is the metric of the unit 2-sphere. It is instructive to construct the well understood conformal transformation onto the Einstein cylinder, described in detail in Frauendiener (2000). The standard visualisation compactifies on null coordinates, so we introduce

$$u = t - r \,, \qquad v = t + r \,,$$

subject to $v - u \geq 0$, and using this coordinate transformation gives the line element

$$\tilde{g} = \mathrm{d}u\mathrm{d}v - (v-u)^2\mathrm{d}\sigma^2 \,.$$

Now, we compactify these coordinates via

$$u = \tan U\,, \qquad v = \tan V\,,$$

which both range over the interval $(-\pi/2, \pi/2)$ with the restriction $V - U \geq 0$. Written in terms of (t, r), the combined coordinate transformation is given by

$$t = \frac{\tan V + \tan U}{2}\,, \qquad r = \frac{\tan V - \tan U}{2}\,.$$

In terms of (U,V), the line element takes the form

$$\tilde{g} = \frac{1}{4\cos^2 U \cos^2 V}\left[4\mathrm{d}U\mathrm{d}V - \sin^2(V-U)\mathrm{d}\sigma^2\right]\,.$$

and in defining the conformal factor $\Omega = 2\cos U \cos V$, we obtain the unphysical metric

$$g = \Omega^2\tilde{g} = 4\mathrm{d}U\mathrm{d}V - \sin^2(V-U)\mathrm{d}\sigma^2\,, \tag{46}$$

which allows for a smooth extension of the spacetime beyond the interval $(-\pi/2, \pi/2)$ in (U, V). We find this extension to be identical to the Einstein cylinder spacetime, which we can see by defining appropriate coordinates; letting $T = V + U$ and $R = V - U$, the line element reads

$$g = \mathrm{d}T^2 - \mathrm{d}R^2 - \sin^2 R\mathrm{d}\sigma^2\,. \tag{47}$$

We can write the direct coordinate transformation as follows

$$t = \frac{\tan\left(\frac{T+R}{2}\right) + \tan\left(\frac{T-R}{2}\right)}{2}\,, \qquad r = \frac{\tan\left(\frac{T+R}{2}\right) - \tan\left(\frac{T-R}{2}\right)}{2}\,. \tag{48}$$

The metric (47) realises Minkowski space as being conformally embedded in the Einstein cylinder. The boundary, $\mathscr{I} = \{\Omega = 0\}$, is given by the transformed physical range of (U, V), which is given as $0 \leq R$, $-\pi \leq R + T \leq \pi$, $-\pi \leq T - R \leq \pi$.

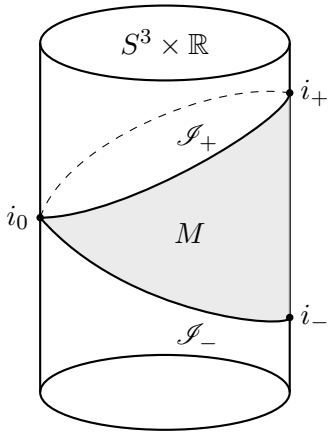


Figure 1: Figure showing the conformal embedding of Minkowski space into the Einstein cylinder. Time coordinate T runs along the height of cylinder, while R goes around the cylinder and angular coordinates (θ, ϕ) are suppressed. Null, spatial and timelike infinite boundaries can be seen.

Figure 1 gives a visual representation with the angular coordinates, (θ, ϕ), suppressed allowing each point to represent a 2-sphere; the highlighted section of the half cylinder gives the embedding described. Extending the spacetime by allowing for negative R values gives an embedding into the section of the full cylinder indicated by the dashed line.

We will restrict our attention to timelike conformal geodesics that propagate from the spacelike hypersurface $\underline{S} = \{t = 0\}$. We will further assume spherical symmetry and take θ and ϕ to be constant, fixed at $\theta = \pi/2$ and $\phi = 0$. Denote by r_* the coordinate r restricted to the hypersurface $\underline{S}$; by varying r_* we can construct a congruence of conformal geodesics. The coordinate R_* can be defined analogously and since $\underline{S} = \{T = 0\}$, we can use (48) to find $R_* = 2 \arctan r_*$. Given that we normalise the tangent vector of the conformal geodesics with respect to the conformal metric for some suitable initial conformal factor, we can define a CGG from this data where the congruence is smooth and caustic free. Left is to choose data for the conformal factor, $\underline{\Theta}$, and one-form, $\underline{b}_a$.

3.1.1 Test data

We first consider two sets of test initial conditions, which are not related to the construction of a CGG. The test initial conditions are summarised with respect to a (T, R) coordinate basis: for both sets we have

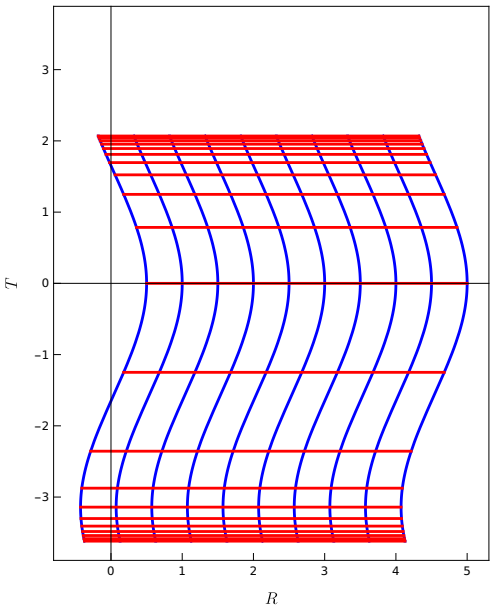
- $\underline{x}^\mu = (0, R_*, \frac{\pi}{2}, 0)$.
- $\underline{u}^\mu = (1, 0, 0, 0)$.

Then $\underline{b}_\mu = (1, -1, 0, 0)$, and $\underline{b}_\mu = (0, -1, 0, 0)$ for the first and second plots respectively where the computations are performed with the unphysical metric, (47). Then, solving (35), we find a congruence, $x^\mu(\tau, \underline{x}^\mu) = (T(\tau, \underline{x}^\mu), R(\tau, \underline{x}^\mu), \pi/2, 0)$, given in Figure 2. We can see in Figures 2a, 2b that the conformal geodesics stay evenly spaced and caustic free, which is necessary the CGG to stay well-defined. Inspecting the level sets of the parameter, τ, we find that the distance travelled between intervals of τ becomes shorter as we increase the parameter, and it appears to asymptote. This is a potential problem as it may happen that the natural parameter asymptotes before it reaches the conformal boundary. In Figure 2a we notice that the level sets are not symmetric, and this behaviour was observed to occur when $\underline{b}_c \underline{u}^c \neq 0$. When looking at different initial data, so the curves stay time symmetric it will henceforth generally be imposed that $\underline{b}_c \underline{u}^c = 0$.

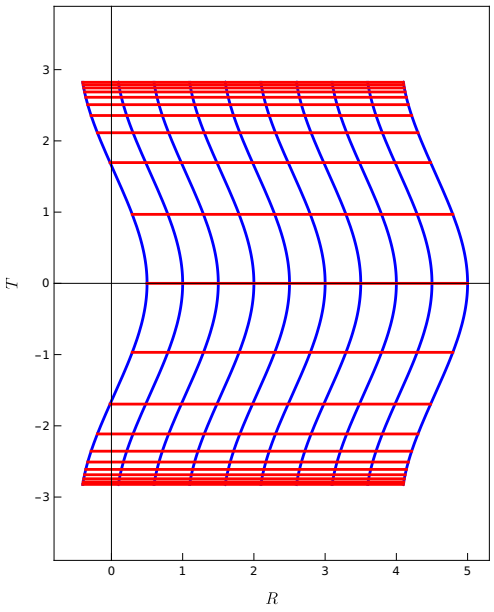
3.1.2 Friedrich's data

We now solve the CGE with the initial data in (Friedrich, 2003). These can be summarised in the (T, R) coordinate basis as follows:

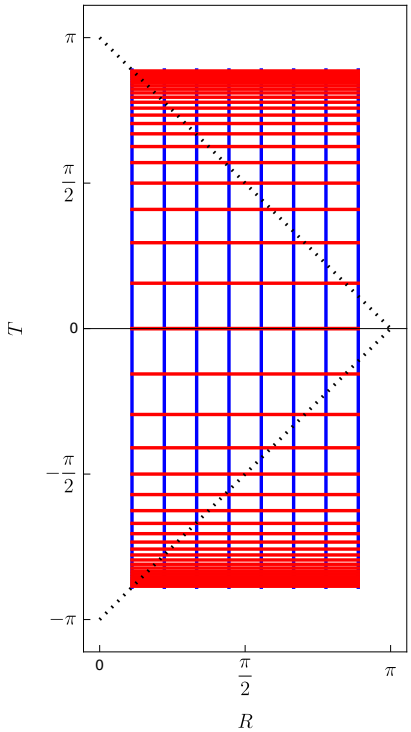
- $\underline{x}^\mu = (0, R_*, \frac{\pi}{2}, 0)$.
- $\underline{\Theta} = 1 + \cos R_*$.
- $\underline{u}^\mu = (1, 0, 0, 0)$.
- $\underline{b}_\mu = (0, \frac{\sin R_*}{1 + \cos R_*}, 0, 0)$.



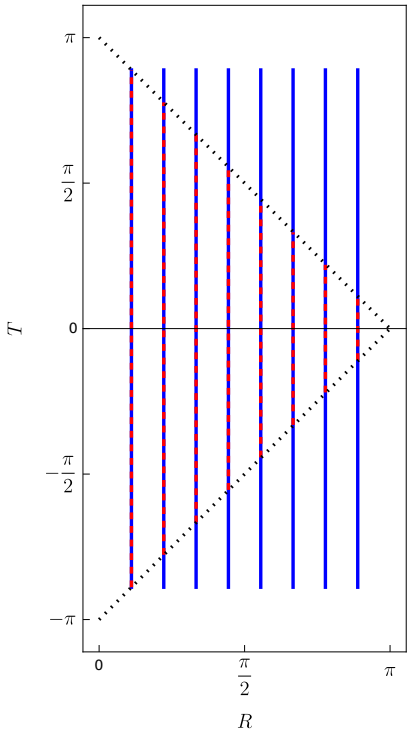
(a) Congruence for first set of test data in (T, R) plane.



(b) Congruence for second set of test data in (T, R) plane.



(c) Congruence for Friedrich's initial data in (T, R) plane.



(d) Congruence for Friedrich's initial data showcasing the analytic solution in red.

Figure 2: Figures showing the congruence of conformal geodesics in blue and level sets in red, solved numerically in Minkowski space as described in text.

The computations are performed with the unphysical metric, (47). These choices for $\underline{x}^\mu$ and $\underline{u}^\mu$ ensure that the conformal geodesics are evolving from the $t = 0$ hypersurface and begin initially tangent while the tangent vector is normalised with respect to the conformal metric defined by the initial conformal factor. The initial conformal factor is chosen to realise the standard embedding into the Einstein cylinder on the $t = 0$ hypersurface. Finally $\underline{b}_a$ is chosen so that the $\underline{\Theta}^{-1}\nabla_a\underline{\Theta} = \underline{b}_a$. The congruence from the above set of initial conditions is plotted in the (T, R) plane in Figure 2. It has been numerically computed with given initial conditions up to $\tau = \frac{2}{r_{*,1}}$, where $r_{*,1}$ denotes the smallest value in the congruence of r_* values considered. Friedrich (2003) gives[3] the analytic solution to the CGE for the above initial conditions as

$$t(\tau) = \frac{2(1 + r_*)}{4 - \tau^2 r_*^2}\,, \qquad r(\tau) = \frac{r_*(4 + \tau^2)}{4 - \tau^2 r_*^2}\,.$$

This solution which is expected to asymptote towards $\mathscr{I}$ as $\tau \to \frac{2}{r_*}$ can be compared with the numerically computed solution. Figure 2d shows that the computed solutions agree with the analytic ones in the physical domain. As seen in Figure 2c, the numerical solution, which is computed in the unphysical domain, indeed smoothly passes through $\mathscr{I}$. Moreover, Figure 2c shows that as $\tau \to \infty$, the conformal geodesics with $R_* = 0$ approach timelike infinity. These behaviours are reflected by the level sets which match with our expectations. These agreements with analytical results give confidence in the numerical computations.

Given that the numerical implementation on Minkowski was successful, we can consider the solutions given in Friedrich (2003) on the more general Schwarzschild spacetime and look at the effects of tweaking the initial data.

3.2 Schwarzchild Metric

We would now like to consider the CGE with the Schwarzschild metric, describing the space around a non-rotating, chargeless mass. In Kruskal-Szekeres[4] coordinates (s, ρ, θ, ϕ), the metric takes the form

$$\tilde{g} = \frac{32m^3 e^{-\frac{r}{2m}}}{r}(\mathrm{d}s^2 - \mathrm{d}\rho^2) - r^2\mathrm{d}\sigma^2\,, \tag{49}$$

where $\mathrm{d}\sigma^2$ is the metric of the unit 2-sphere and r is defined implicitly via the equation

$$s^2 - \rho^2 = \left(1 - \frac{r}{2m}\right)\exp\left(\frac{r}{2m}\right),$$

or explicitly using the Lambert W function[5] on the principal branch, W_0, as

$$r = 2m\left(W_0\left(\frac{\rho^2 - s^2}{e}\right) + 1\right).$$

[3]Note that the text reads $r = \frac{r_*^2(4+\tau^2)}{4-\tau^2 r_*^2}$ but this is a typo.

[4]We will use non-standard notation for the coordinates in order to stay in line with Minkowski, where capital letters are denoting compactified coordinates.

[5]Defined as the multivalued inverse function of the complex function $f(x) = xe^x$. For real numbers two branches suffice, denoted $W_0(z)$ and $W_1(z)$ respectively. Where $f(x) \geq 0$, we have $x = W_0(xe^x)$.

These coordinates can be defined from the standard Schwarzschild coordinates, (t, r, θ, ϕ), via the transformations

$$s = \left(\frac{r}{2m} - 1\right)^{\frac{1}{2}} \exp\left(\frac{r}{4m}\right) \sinh\left(\frac{t}{4m}\right) ,$$
$$\rho = \left(\frac{r}{2m} - 1\right)^{\frac{1}{2}} \exp\left(\frac{r}{4m}\right) \cosh\left(\frac{t}{4m}\right) ,$$

and they extend analytically from the Schwarzschild patch to the region $-\infty < s^2 - \rho^2 < 1$, where $-\infty < \rho < \infty$. To view the congruence on the Penrose diagram for Schwarzschild, we follow a similar procedure as in Section 3.1. First, we define the compactified coordinates (S, P) via

$$s = \frac{\tan(S+P) + \tan(S-P)}{2} , \qquad \rho = \frac{\tan(S+P) - \tan(S-P)}{2} ,$$

which renders the physical metric as

$$\tilde{g} = \frac{32m^3 e^{-\frac{r}{2m}}}{r\cos^2(S-P)\cos^2(P+S)} \left(\mathrm{d}S^2 - \mathrm{d}P^2\right) - r^2 \mathrm{d}\sigma^2 ,$$

where r is now defined by the equation

$$r = 2m\left(W_0\left(-\frac{\tan(S-P)\tan(P+S)}{e}\right) + 1\right) .$$

By taking the conformal factor $\Omega = \cos(S-P)\cos(P+S)$ we can write the unphysical metric

$$g = \Omega^2 \tilde{g} = \frac{32m^3 e^{-\frac{r}{2m}}}{r} \left(\mathrm{d}S^2 - \mathrm{d}P^2\right) - r^2 \cos^2(S-P)\cos^2(P+S)\mathrm{d}\sigma^2 . \tag{50}$$

Unfortunately, although formally well defined at infinity, this metric was not found to be suitable for evolving conformal geodesics through $\mathscr{I}$ so we will find it useful to define other coordinates. The retarded and advanced Eddington-Finkelstein coordinates are defined respectively as

$$u = t - (r + 2m\log(r-2m)) , \qquad v = t + (r + 2m\log(r-2m)) ,$$

from the standard Schwarzschild coordinates (t, r, θ, ϕ). In these coordinates the metric takes the forms

$$\tilde{g}_u = \left(1 - \frac{2m}{r}\right)\mathrm{d}u^2 + 2\mathrm{d}u\mathrm{d}r - r^2\mathrm{d}\sigma^2 , \qquad \tilde{g}_v = \left(1 - \frac{2m}{r}\right)\mathrm{d}v^2 - 2\mathrm{d}v\mathrm{d}r - r^2\mathrm{d}\sigma^2 .$$

Considering the retarded metric, and using the coordinate $z = \frac{1}{r}$ we obtain the unphysical metric via the conformal factor $\Omega = z$, in the forms

$$g_u = \Omega^2\tilde{g}_u = z^2(1-2mz)\mathrm{d}u^2 - 2\mathrm{d}u\mathrm{d}z - \mathrm{d}\sigma^2 , \quad g_v = \Omega^2\tilde{g}_v = z^2(1-2mz)\mathrm{d}v^2 + 2\mathrm{d}v\mathrm{d}z - \mathrm{d}\sigma^2 . \tag{51}$$

These allow analytic extensions of the conformal geodesics through $\mathscr{I}^+$ and $\mathscr{I}^-$ respectively, where the set $\{z = 0\}$ represents $\mathscr{I}$.

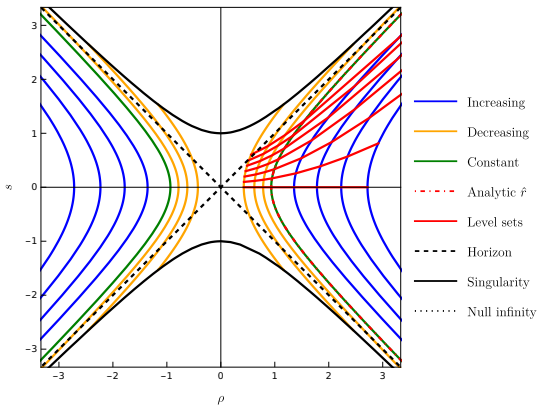
As in Section 3.1, we restrict our attention to timelike conformal geodesics that propagate from the spacelike hypersurface $\underline{S} = \{t = 0\}$. Denote by r_* the Schwarzschild radius, r, restricted to this hypersurface; by varying r_* we can construct a congruence of conformal geodesics. Assuming spherical symmetry, θ and ϕ are taken to be constant, fixed at $\theta = \pi/2$ and $\phi = 0$. Given that we normalise the tangent vector of the conformal geodesics with respect to the conformal metric for some suitable initial conformal factor, we can define a CGG from this data where the congruence is smooth and caustic free. We take $m = \frac{1}{2}$ which puts the Schwarzschild horizon at $r = 1$ for numerical calculation but will not affect the qualitative behaviour.

3.2.1 Friedrich's data on maximal Kruskal-Szekeres spacetime

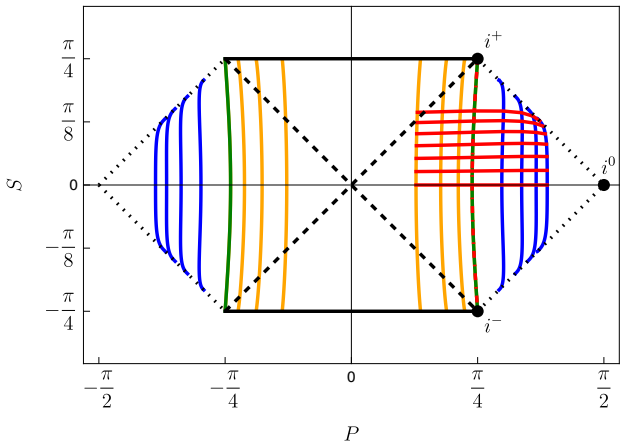
To start, we solve the CGE with the initial data given in Friedrich (2003), which can be summarised in the (s, ρ) coordinate basis as:

- $\underline{x}^\mu = (0, e^{\frac{r_*}{4m}}\sqrt{\frac{r_*}{2m} - 1}, \frac{\pi}{2}, 0)$, where r_* is taken at different values to form a congruence.
- $\underline{u}^\mu = (\sqrt{\frac{r_*^5 e^{\frac{r_*}{2m}}}{32m^3}}, 0, 0, 0)$.
- $\underline{b}_\mu = (0, -\frac{16m^2 e^{-\frac{r_*}{4m}}\sqrt{\frac{r_*}{2m}-1}}{r_*^2}, 0, 0)$.
- $\underline{\Theta} = \frac{1}{r_*^2}$.

The computations are performed with the physical metric, (49). These conditions ensure that the conformal geodesics are evolving from the $t = 0$ hypersurface and begin initially tangent, normalised with respect to the conformal metric defined by the initial conformal factor. Figure 3a shows the congruence plotted in the (s, ρ) plane. For clarity, the



(a) Figure showing the congruence of conformal geodesics in Schwarzschild space on the (s, ρ) plane with initial data as in text. Level sets are printed for the first quadrant only.



(b) Figure showing the congruence of conformal geodesics in Schwarzschild space on the (s, ρ) plane with initial data as in text. Timelike and spatial infinities are highlighted.

level sets are only printed for the first quadrant but the initial data chosen ensures that

the level sets are symmetric under the symmetries $s \to -s$ and $\rho \to -\rho$. Friedrich (2003) derives the behaviour of the conformal geodesics with the above data, showing that there exists a critical value of r_*, denoted $\hat{r}$, that divides the congruence according to their asymptotic behaviour. Above the critical value, conformal geodesics pass through $\mathscr{I}^{\pm}$ as $\tau \to \pm\tau_{scri} \equiv (r_*^2 - 2mr_*))^{-\frac{1}{2}}$, and below it, conformal geodesics move towards the singularity. On the critical value of r_*, the conformal geodesic approaches time-like infinity as $\tau \to \tau_i \equiv \frac{2}{\sqrt{5}m}$. The one-form can be thought of as giving an acceleration that acts against the 'pull' of the Schwarzschild singularity that decays as we increase r. Then it is natural that we have a radially constant conformal geodesic if there is a point where these effects cancel. The analytic solution for the conformal geodesic on the critical value $\hat{r}$ can be written in standard Schwarzschild coordinates as

$$t(\tau) = \frac{25}{4} m \log\left(\frac{1+\sqrt{r_*^2 - 2mr_*^2}\tau}{1-\sqrt{r_*^2 - 2mr_*^2}\tau}\right), \qquad r(\tau) = \hat{r} \equiv \frac{5}{2}m\,.$$

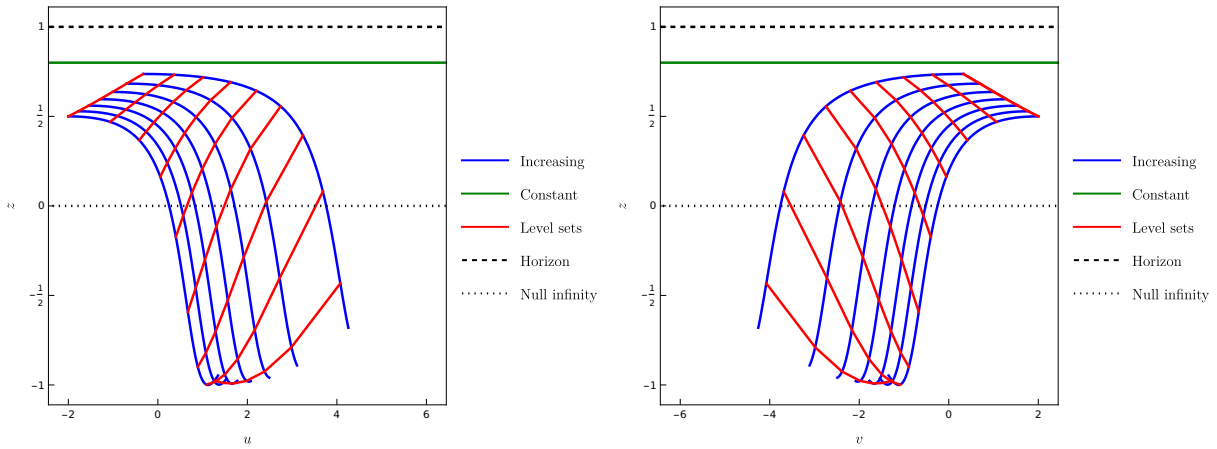
The numerical solution, plotted in Figure 3a, is shown to exhibit all of these listed behaviours. These curves were then overlaid onto the standard conformal Penrose diagram for Schwarzschild in the (S, P) plane in Figure 3b. As we can see, the congruence is not smoothly evolving through $\mathscr{I}$. In order to observe this smooth evolution, we need to consider the different regions separately. Also, notice that the level sets intersect with other points of $\mathscr{I}$ long before we reach i^+. This behaviour can be changed with different initial data.

3.2.2 Friedrich's data in Eddington-Finkelstein coordinates adapted to $\mathscr{I}^{\pm}$

Now, we solve the CGE with the intent of numerically evolving them through $\mathscr{I}$. We use the same set of initial conditions as in Section 3.2.1, and solve in the unphysical retarded/advanced metrics, (51). The transformed initial conditions for unphysical metric in an inverted retarded/advanced coordinate basis, $(u/v, z, \theta, \phi)$, are summarised:

- $\underline{x}^{\mu} = \left(-r_* \pm 2m\log(r_* - 2m), \frac{1}{r_*}, \frac{\pi}{2}, 0\right)$, where the plus and minus correspond to the advanced and retarded coordinate respectively.
- $\underline{u}^{\mu} = (\frac{1}{\sqrt{1-\frac{1}{r_*}}}, 0, 0, 0)$
- $\underline{b}_{\mu} = (0, r_*, 0, 0)$
- $\underline{\Theta} = \frac{1}{r_*^2}$.

The congruence for these sets of initial data can be seen in the (T, R) plane in Figure 4. As can be seen, the curves pass smoothly through the $z = 0$ line representing $\mathscr{I}^+$ and $\mathscr{I}^-$ for the retarded and advanced coordinate systems respectively. For numerical simulations, it is important that the curves remain caustic free just outside $\mathscr{I}$ as the numerical stepping may enter this domain. We observe that the curves begin to bend and eventually intersect after they cross this line but remain caustic free for a small period.



(a) Congruence for the retarded coordinate spacetime in (u, z) plane.

(b) Congruence for the advanced coordinate spacetime in (v, z) plane.

Figure 4: Figures showing the congruence of conformal geodesics in Schwarzschild Space computed in unphysical metrics adapted to $\mathscr{I}^{\pm}$. Level sets of the parameter are printed in red.

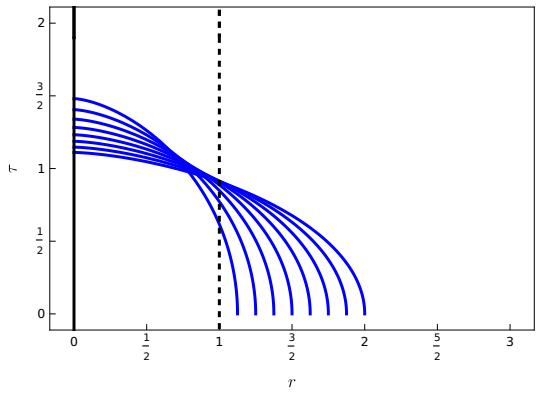
3.2.3 Alternate initial data

Next, we are looking to investigate different choices of initial data that respect the construction of the CGG and also have $\underline{b}_c \underline{u}^c = 0$. This second condition simplifies analytic expressions given in Section B and ensures that the curves are symmetric about time reflection. We still take $\underline{\Theta} = \frac{1}{r_*^2}$, and give a list of the alternate initial data for $\underline{b}$ in a (t, r) coordinate basis:

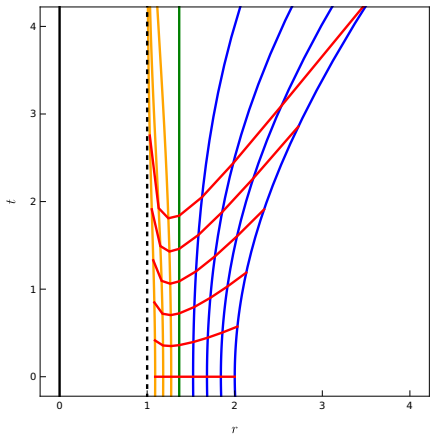
- $\underline{b}_\mu = (0, 0, 0, 0)$.
- $\underline{b}_\mu = (0, 1, 0, 0)$.
- $\underline{b}_\mu = (0, \frac{-1}{r_*^2\sqrt{1-\frac{2m}{r_*}}}, 0, 0)$.

In this section, all conformal geodesics are computed with the physical metric, (49). Figure 5b gives the congruence in the (t, r) plane with the initial data as before but taking $\underline{b}_c$ to be constant. By comparing to the Figure 5a which plots the congruence with a zero one-form, we see that the one form indeed behaves like an acceleration. We can also view the constant initial one-form congruence in the (S, P) plane in Figure 5c, which is seen to be qualitatively similar to the congruence generated by the data in Friedrich (2003); that is caustic free and exhibiting the same level set behaviour. Where (B.15) had only one solution, corresponding to constant r_*, we generally found the curves to be caustic free given the one form was of the correct sign so that the curves did not bend in towards the constant r_* curve.

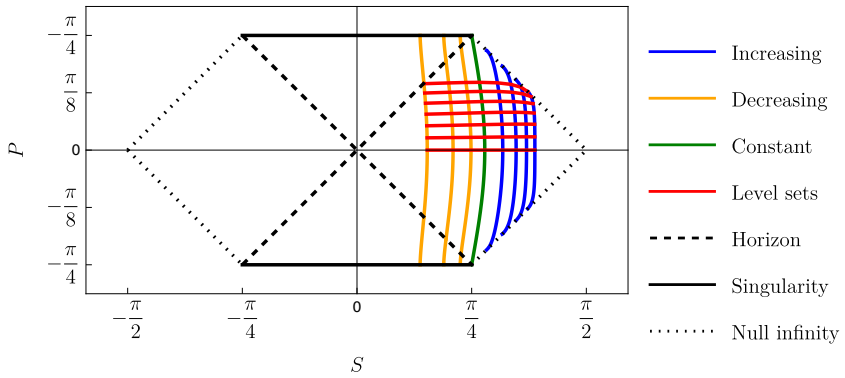
We can also construct a set of curves such that the parameter value to reach $\mathscr{I}^{\pm}$, $\tau_{\pm}$, is the same for all conformal geodesics. Using the general formula for $\tau_{\pm}$, given in (B.7) this value is constant where $\beta = c\Theta$ for some $c \in \mathcal{R}^+$. Figures 5d and 5e give the congruence for $c = 1$ in the (t, r) and (S, P) planes respectively.



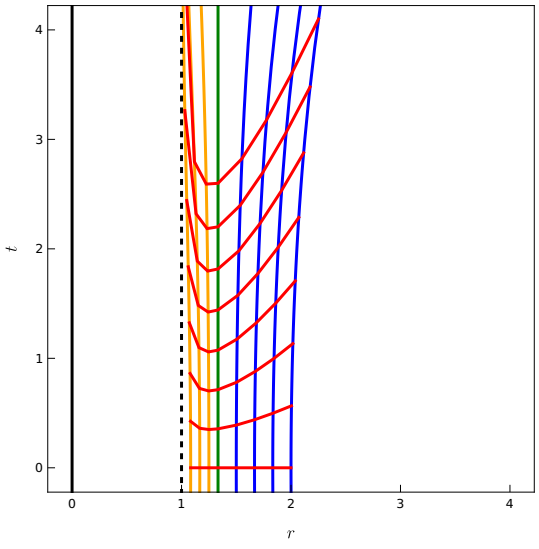
(a) Congruence for the zero one-form initial data (metric geodesics) plotted in (τ, r) space.



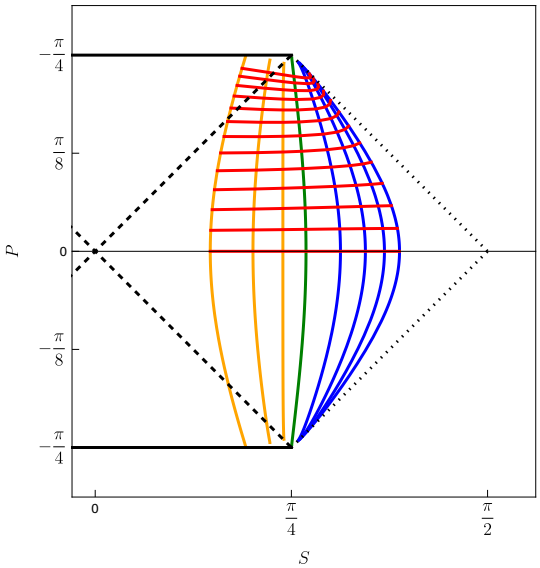
(b) Congruence for the constant one-form initial data in (t, r) space.



(c) Congruence for the constant one-form initial data in (S, P) space.



(d) Congruence in (t, r) plane for the constant $\tau_\pm$ initial data.

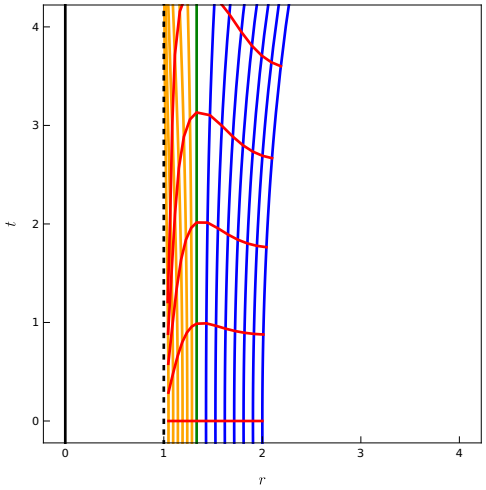


(e) Congruence in (S, P) plane for the constant $\tau_\pm$ initial data.

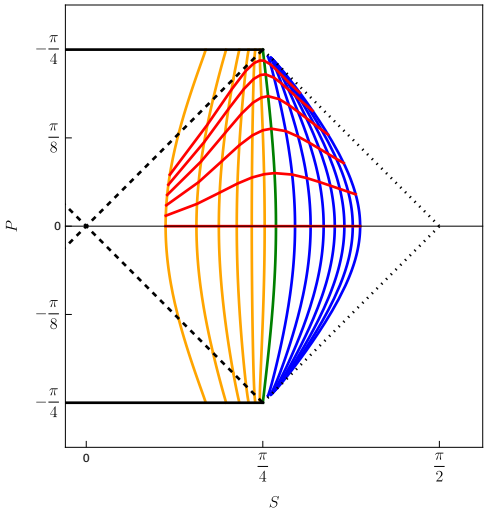
Figure 5: Figures showing the congruence of conformal geodesics in Schwarzschild space for various initial data as described in text. A legend that applies to all diagrams is visible in the centre image.

3.2.4 Initial data adapted to $i^{\pm}$

With the intent of finding level sets that are adapted to $i^{\pm}$, we require that $\tau_{\pm}(\hat{r}) < \tau_{\pm}(r_*), \tau_0(r_*)$ for all $r_* > \frac{m}{2}$, where τ_0 denotes the parameter value to reach the singularity, given by (B.16) evaluated at $r = 0$, and β is such that $\hat{r}$ is the unique solution to (B.15). We can achieve this given we change the conformal factor but we must make sure $\underline{\Theta}$ defines a compactification on the initial time slice. We require that the initial conformal factor is $O(1/r)$ as $r \to \infty$, which is sufficient (Frauendiener, 2000). The numerically calibrated pair $\underline{b}_\mu = (0, -\frac{1}{r_*^2\sqrt{1-2m/r_*}}, 0, 0)$, and $\underline{\Theta} = \frac{(r_*-1.1)^2+0.08455}{2r_*^3 \log r_*}$, are a suitable choice to achieve this behaviour and Figure 6 gives the resulting congruence in the (t, r) and (S, P) planes. The level sets in Figure 6b can be seen curving towards future timelike infinity as one would expect.



(a) Congruence for the $i_{\pm}$ adapted initial data in (t, r) space.



(b) Congruence for the $i_{\pm}$ adapted initial data in (S, P) space.

Figure 6: Figures showing the congruence of conformal geodesics in Schwarzschild Space. The legend follows conventions as in previous figures

3.3 Kerr Metric

The Kerr metric describes the vacuum space around a rotating, chargeless black hole. It can be considered a generalisation of the Schwarzschild metric when it is describing a black hole, as we may now consider the effects of rotation. In standard Boyer-Lindquist coordinates, (t, r, θ, ϕ), the physical metric takes the form

$$\tilde{g} = \left(1 - \frac{2mr}{\Sigma}\right) \mathrm{d}t^2 - \frac{\Sigma}{\Delta}\mathrm{d}r^2 - \Sigma \mathrm{d}\theta^2 - \left(r^2 + a^2 + \frac{2mra^2}{\Sigma}\sin^2\theta\right)\sin^2\theta \mathrm{d}\phi^2 + \frac{4mar\sin^2\theta}{\Sigma}\mathrm{d}t\mathrm{d}\phi\,, \tag{52}$$

where for brevity we have $\Sigma \coloneqq r^2 + a^2\cos^2\theta$, $\Delta \coloneqq r^2 - 2mr + a^2$, and the constant $a \coloneqq \frac{J}{m}$, where J represents the total angular momentum (Townsend, 1997). The metric (52) reduces to Schwarzschild metric in Schwarzschild coordinates as $a \to 0$ and so can naturally be used to extend the initial data in Friedrich (2003). In order to extend the

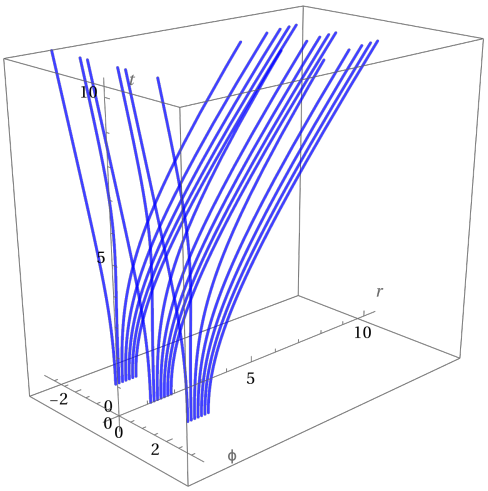
spacetime through the horizons we introduce Kerr coordinates, (v, r, θ, χ) as in Townsend (1997), which are analogous to Eddington-Finkelstein coordinates on Schwarzschild. Now working in Boyer-Lindquist coordinates we take $\underline{S} = \{t = 0\}$ and consider conformal geodesics in axisymmetry where θ is fixed at $\theta = \pi/2$. We denote by r_* and ϕ_*, the coordinates restricted to this hypersurface. For numerical evaluations we take $m = \frac{1}{2}$, and $a = \frac{1}{4}$. It is generally considered unphysical to have $a > m$ and we take this approach in the choice of a (Thorne et al., 2000). This puts the outer horizon at $r = r_H^+ := \frac{2+\sqrt{3}}{4}$.

3.3.1 Friedrich's data

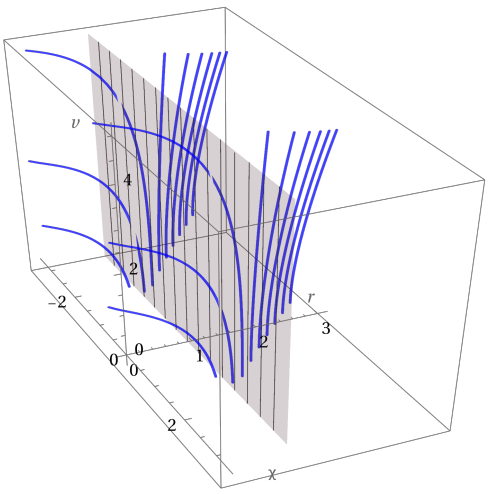
We introduce the obvious generalisation by replacing the Schwarzschild coordinate with the Boyer-Lindquist one in Friedrich's data. Then in a (t, r, θ, ϕ) coordinate basis we have

- $\underline{x}^\mu = (0, r_*, \frac{\pi}{2}, \phi_*)$, where r_*, ϕ_* are taken at different values to form a congruence.
- $\underline{u}^\mu = (r_*^2\sqrt{\frac{r_*}{r_*-2m}}, 0, 0, 0)$.
- $\underline{b}_\mu = (0, -\frac{2}{r_*}, 0, 0)$.
- $\underline{\Theta} = \frac{1}{r_*^2}$.

The curves are computed numerically in (v, r, θ, χ) coordinates with the corresponding physical metric so that we may evolve through the coordinate singularity. Figures 7a and 7b give the congruence for the above initial data in the (t, r, ϕ) plane and (v, r, χ) planes respectively. For visibility we only plot a few values of r_*, ϕ_*. Figure 7a also shows, similar to Schwarzschild, there is some critical $r_* = \hat{r}$ where for $r_* > \hat{r}$ conformal geodesics move towards $\mathscr{I}$ and for $r_* < \hat{r}$ they move towards the singularity. As r_* approaches the singularity the curves rotate more in ϕ but they appear to stay regular and caustic free, at least locally.



(a) Congruence for the Kerr spacetime in (t, r, ϕ) space.



(b) Congruence for the Kerr spacetime in (v, r, χ) space. Overlaid is the outer horizon surface.

Figure 7: Figures showing the congruence of conformal geodesics in Kerr Space, with initial data as described in the text.

4 Conclusions

In this report we saw that conformal geodesics are a useful structure in analysis of the conformal compactification of vacuum spacetimes. An important application is that they are used in the construction of the conformal Gauss gauge, which fixes the gauge for the generalised conformal field equations by evolving data given on an initial hypersurface along conformal geodesics. A better understanding of conformal geodesics tells us about the constructed conformal Gauss gauge. As mentioned, this gauge been used in numerical simulations of perturbations on the Schwarzschild spacetime (Frauendiener & Stevens, 2021) to extract information 'at infinity'. The conformal geodesic equations were solved numerically for a number of different cases; we confirmed analytic results for the initial data as given in Friedrich (2003), provided visualisation that doesn't exist elsewhere, and explored different gauge choices. In particular, initial data was shown to exist that exhibits different behaviour in the level sets than to Friedrich's choice, that is potentially better suited to investigations of $i^{\pm}$. On the Kerr spacetime, we were able to generalise Friedrich's initial data and find how the curves evolved, showing that they are caustic free locally. These curves could be used to form a conformal Gauss gauge on perturbations of the Kerr spacetimes for use in numerical simulations.

Future work would include a more detailed analysis of the Kerr spacetime. The curves computed seem to indicate that the congruence is caustic free globally, especially as the conformal geodesics approach those computed in Schwarzschild as $r \to \infty$ but a global computation is desirable. The basic approach would be similar to Schwarzschild: we can utilise double-null Eddington-Finkelstein like coordinates, (u, v, θ, χ), where v, χ are defined as in Section 3.3 and $u = t - \int \frac{r^2+a^2}{\Delta}$, which then allow for a transformation into Kruskal-Szekeres like coordinates (U, V, θ, χ) which would cover the maximally extended Kerr spacetime. It would then be possible to compactify these coordinates and investigate the global structure visually. Also necessary would be to vary the fixed θ, as the behaviour of the conformal structure in the Kerr spacetime depends on θ in a way that Schwarzschild doesn't; this is evidenced by the fact that the curvature singularity is only present for $\theta = \pi/2$. Further, we would like to investigate the behaviour of the level sets of the parameter for different initial data, as in Schwarzschild.

We could also investigate the congruence in the more general Kerr-Newman metric which models the vacuum space around a charged, rotating black hole. This would be a straightforward generalisation where we could use very similar methods as in studying the Kerr metric. We would not expect the behaviour to be very different in this case.

A different approach would be to introduce non-vacuum spacetimes that have sufficient matter fall-off conditions to be considered asymptotically flat. The GCFE and CGG can be generalised for matter, see for example Lübbe and Kroon (2013), so the methods applied in this report could be used as a basis for non-vacuum spacetimes as well.

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Appendix A Conformal Curvature Quantities

In order to understand how the Riemann curvature tensor changes under a conformal rescaling we will find it easiest to decompose $R_{ab}{}^{cd}$ as follows

$$R_{ab}{}^{cd} = C_{ab}{}^{cd} + 4P_{[a}{}^{[c}\delta_{b]}^{d]}\,, \tag{A.1}$$

where $C^{a}{}_{bcd}$ is the Weyl tensor which is defined as the totally trace-free Riemann tensor, and P_{ab} is the Schouten tensor that can be defined from the Ricci tensor by

$$P_{ab} \coloneqq \frac{1}{2}\left(R_{ab} - \frac{1}{6}g_{ab}R\right)\,. \tag{A.2}$$

Next we derive some useful identities for various curvature quantities. Consider the second Bianchi identity for the Riemann tensor which reads $\nabla_{[e}R_{ab]cd} = 0$; from this, we can insert (A.1) (raising/lowering the indices appropriately) to find

$$\nabla_{[e}R_{ab]c}{}^{d} = \nabla_{[e}C_{ab]c}{}^{d} + 2\nabla_{[e}\left(P_{a|c|}\delta_{b]}^{d} - P_{a}{}^{d}g_{b]c}\right) = 0\,. \tag{A.3}$$

Then we can take a contraction of (A.3), utilising the identity $\nabla_{[e}C_{ab]c}{}^{e} = 1/3\nabla_{e}C_{abc}{}^{e}$, to give

$$\nabla_{e}C_{abc}{}^{e} + 2\nabla_{[a}P_{b]c} + 2g_{bc}\nabla_{[a}P_{e]}{}^{e} - 2g_{ac}\nabla_{[b}P_{e]}{}^{e} = 0\,, \tag{A.4}$$

a further trace on b, c gives us the equation

$$\nabla_{[a}P_{e]}{}^{e} = 0\,, \tag{A.5}$$

as the Weyl tensor is trace-free. Inserting (A.5) into (A.4) allows us to see

$$\nabla_{e}C_{abc}{}^{e} + 2\nabla_{[a}P_{b]c} = 0\,,$$

which is equivalent to

$$\nabla_{a}C^{a}{}_{bcd} + 2\nabla_{[d}P_{c]b} = 0\,. \tag{A.6}$$

Finally, consider the contracted Bianchi identity which is given as $\nabla^{b}R_{ab} = \frac{1}{2}\nabla_{a}R$; inserting (A.2) gives

$$\nabla^{b}P_{ab} = \frac{1}{6}\nabla_{a}R\,. \tag{A.7}$$

Now, given a conformal rescaling of a some metric $\tilde{g}_{ab} \to g_{ab} \coloneqq \theta^{2}\tilde{g}_{ab}$ let us define

$$\Upsilon_{a} = \theta^{-1}\nabla_{a}\theta\,, \tag{A.8}$$

where the connection can be taken to be the Levi-Civita connection for any metric in the conformal class, as they act the same on scalar fields. Next, define

$$\Upsilon_{ab}{}^{c} = \delta_{a}^{c}\Upsilon_{b} + \delta_{b}^{c}\Upsilon_{a} - g_{ab}g^{cd}\Upsilon_{d}\,, \tag{A.9}$$

noting that $\Upsilon_{ab}{}^{c} = \Upsilon_{ba}{}^{c}$. As $g^{ab} = \theta^{-2}\tilde{g}^{ab}$, the quantity $g_{ab}g^{cd}$ is invariant under conformal transformations, and can be evaluated by any metric in the conformal class. We can then write the transformation of various curvature quantities under the conformal rescaling. Curvature quantities defined with respect to $\tilde{g}_{ab}$ will be denoted with a tilde, and are

otherwise are assumed to be defined with respect to g_{ab}. We find, for the Levi-Civita connection of the new metric acting on arbitrary vector field, v^a,

$$\nabla_a v^b = \tilde{\nabla}_a v^b + \Upsilon_{ac}{}^b v^c \,, \tag{A.10}$$

where the usual Leibniz property can be used to extract the action on tensor fields of any valence. Note that in the following, indices can be assumed to be raised/lowered with g_{ab}. We can write (A.1) equivalently as

$$R^a{}_{bcd} = C^a{}_{bcd} + 2\left(\delta^a_{[c} P_{d]b} - g_{b[c} P_{d]}{}^a\right) , \tag{A.11}$$

or

$$R_{abc}{}^d = C_{abc}{}^d - 2\left(\delta^d_{[a} P_{b]c} - g_{c[a} P_{b]}{}^d\right) . \tag{A.12}$$

It will also be helpful to contract (A.11) on a, c to find

$$R_{ab} = 2P_{ab} + g_{ab} P \,, \tag{A.13}$$

for $P = P_c{}^c = R/6$. It turns out that the Weyl tensor transforms as

$$C^a{}_{bcd} = \tilde{C}^a{}_{bcd} \,,$$

while for the Schouten tensor we find

$$P_{ab} = \tilde{P}_{ab} - \nabla_a \Upsilon_b - \Upsilon_a \Upsilon_b + \frac{1}{2}\tilde{g}_{ab}\tilde{g}^{cd}\Upsilon_c \Upsilon_d \,. \tag{A.14}$$

We can then find the difference in the Riemann tensors, using (A.1) and (A.14), as

$$R_{abc}{}^d = \tilde{R}_{abc}{}^d - 2\left(\nabla_{[a}\Upsilon_{b]c}{}^d + \Upsilon_{c[a}{}^e \Upsilon_{b]e}{}^d\right) . \tag{A.15}$$

From (A.15), it is simple to derive the relationship between Ricci tensors as

$$R_{ab} = \tilde{R}_{ab} - 3\nabla_a \Upsilon_b + \nabla_b \Upsilon_a - 2\Upsilon_a \Upsilon_b + g_{ab}\left(2\Upsilon_c \Upsilon^c - \nabla^d \Upsilon_d\right) , \tag{A.16}$$

which can also be written

$$R_{ab} = \tilde{R}_{ab} - \frac{2}{\theta}\nabla_a \nabla_b \theta + g_{ab}\left(\frac{3}{\theta^2}\nabla_c \theta \nabla^c \theta - \frac{1}{\theta}\nabla^c \nabla_c \theta\right) , \tag{A.17}$$

and, contracting by g_{ab}, the relationship between Ricci scalars as

$$R = \frac{1}{\theta^2}\tilde{R} + 6\left(\Upsilon^a \Upsilon_a - \nabla^a \Upsilon_a\right) , \tag{A.18}$$

where $\tilde{R} = \tilde{g}^{ab}\tilde{R}_{ab} = \theta^2 g^{ab}\tilde{R}_{ab}$.

Appendix B Adapted CGE

We would like to discuss a helpful parametrisation that simplifies the expression of the CGE, (35), with respect to the physical metric, when we assume that the CGG conditions,

(42), hold, and the physical metric is a vacuum spacetime with $\lambda = 0$ (meaning $\tilde{P}_{ab} = 0$). Begin by defining the one-form

$$\hat{b}_a \coloneqq b_a - \frac{b_c u^c}{\tilde{g}_{ef} u^e u^f} \tilde{g}_{ba} u^b \,,$$

then clearly

$$\hat{b}_c u^c = 0 \,, \qquad \hat{b}_e \hat{b}_f \tilde{g}^{ef} = b_e b_f \tilde{g}^{ef} - \frac{(b_c u^c)^2}{\tilde{g}_{ef} u^e u^f} \,. \tag{B.1}$$

Also noticing that, via (38), we have

$$u^c \tilde{\nabla}_c \left(\hat{b}_e \hat{b}_f \tilde{g}^{ef} \right) = 0 \,, \tag{B.2}$$

we define $\beta^2 \coloneqq -\underline{\hat{b}}_e \underline{\hat{b}}_f \tilde{g}^{ef}$ as a constant along conformal geodesics. Note that we assume $\underline{b}_e \underline{b}_f \tilde{g}^{ef} \leq 0$ which allows us to assume $-\hat{b}_e \hat{b}_f \tilde{g}^{ef} \geq 0$. Then, utilising (38), (B.1) and inputting the CGG, (42), we can write the CGE, (35), in the form

$$\Theta u^c \left(\tilde{\nabla}_c \Theta u^a \right) = \tilde{g}^{ac} \hat{b}_c \,, \qquad \Theta u^c \tilde{\nabla}_c \hat{b}_a = -\Theta \hat{b}_e \hat{b}_f \tilde{g}^{ef} \tilde{g}_{ac} u^c = \beta^2 \Theta \tilde{g}_{ac} u^c \,. \tag{B.3}$$

It is natural to introduce the parameter transformation $\tau \to \tilde{\tau}$ as

$$\tilde{\tau}(\tau) = \int_0^\tau \frac{\mathrm{d}\tau'}{\Theta(\tau')} \,, \tag{B.4}$$

in order that $\tilde{u}^a(\tilde{\tau}) = \Theta u^a(\tau)$ satisfies

$$\tilde{g}_{ab} \tilde{u}^a \tilde{u}^b = 1 \,. \tag{B.5}$$

Then (B.3) takes the form

$$\tilde{u}^c \tilde{\nabla}_c \tilde{u}^a = \tilde{g}^{ac} \hat{b}_c \,, \qquad \tilde{u}^c \tilde{\nabla}_c \hat{b}_a = \beta^2 \tilde{g}_{ac} \tilde{u}^c \,, \tag{B.6}$$

which is known as the $\tilde{g}$ adapted form of the CGE (Friedrich, 2003). From (45), we find that the conformal factor, $\Theta(\tau)$, takes zeros at

$$\tau_\pm = 2\underline{\Theta} \frac{\pm|\beta| + \underline{\Theta}(\underline{b}_c \underline{u}^c)}{\beta^2 - \underline{\Theta}^2 (\underline{b}_c \underline{u}^c)^2} = \frac{2\underline{\Theta}}{\pm|\beta| - \underline{\Theta}\,(\underline{b}_c \underline{u}^c)} \,, \tag{B.7}$$

which allows us to write

$$\Theta(\tau) = \frac{\underline{b}_e \underline{b}_f \tilde{g}^{ef}}{4\underline{\Theta}} (\tau - \tau_-)(\tau - \tau_+) \,.$$

Then, via direct integration of (B.4), we see that

$$\tilde{\tau}(\tau) = \frac{1}{|\beta|} \log\left(\frac{\tau_+(\tau - \tau_-)}{\tau_-(\tau - \tau_+)} \right) = \frac{1}{|\beta|} \log\left(\frac{\underline{\Theta}(2 + \underline{b}_c \underline{u}^c) + \tau|\beta|}{\underline{\Theta}(2 + \underline{b}_c \underline{u}^c) - \tau|\beta|} \right) . \tag{B.8}$$

There is some extra reduction in the CGE that we can achieve if we assume the physical metric has a special form. In this derivation, let indices $(A, B, \dots)$ and $(\mathcal{A}, \mathcal{B}, \dots)$

represent the components $(0,1)$ and $(2,3)$ respectively in some coordinate basis. Then, let us assume that the physical metric has the form

$$\tilde{g} = g_{AB}dx^A dx^B + g_{\mathcal{AB}}dx^{\mathcal{A}}dx^{\mathcal{B}}\,, \tag{B.9}$$

where $g_{AB} = g_{AB}(x^A)$ and $\det(g_{AB}) < 0$. Schwarzschild and Minkowski space both have this form. Then the conformal geodesic equations admit *spherically symmetric solutions*, that is take $u^{\mathcal{A}} = 0$, $b_{\mathcal{A}} = 0$; the remaining equations take the same form as (B.6):

$$\left(\tilde{u}^B\tilde{\nabla}_B\tilde{u}\right)^A = \tilde{g}^{AB}\hat{b}_B\,, \qquad \left(\tilde{u}^B\tilde{\nabla}_B\hat{b}\right)_A = \beta^2\tilde{g}_{AB}\tilde{u}^B\,. \tag{B.10}$$

To see this, we can simply compute the full equations with Christoffel symbols based on the given metric form, and take $u^{\mathcal{A}} = 0$, $b_{\mathcal{A}} = 0$, a process that is performed in Friedrich (2003). The final adaption is to notice that these equations actually equivalent so we only need solve one. Because of the equations (42), (B.2) and (B.1) restricted to $A = (0,1)$ we have

$$\tilde{g}_{AB}u^A u^B = \Theta^{-2}\,, \qquad \hat{b}_E\hat{b}_F\tilde{g}^{EF} = \beta^2\,, \qquad \hat{b}_A u^A = \hat{b}_0 u^0 + \hat{b}_1 u^1 = 0\,. \tag{B.11}$$

Then using (B.9) and the conditions (B.11) we have

$$\Theta^2\beta^2\Delta = \left(\frac{\hat{b}_1}{u^0}\right)^2\,,$$

where $\Delta = \det(g_{AB})$. This implies that

$$\hat{b}_A = \pm\sqrt{|\Delta|}\beta\Theta h_{AB}u^B = \pm\sqrt{|\Delta|}\beta h_{AB}\tilde{u}^B\,, \tag{B.12}$$

where we define $h_{AB} = (\mathrm{d}x^0 \wedge \mathrm{d}x^1)_{AB}$ and the sign depends on the initial conditions. From this we can write the second equation in (B.10), letting $\tilde{v}_A = h_{AB}\tilde{u}^B$, as

$$\beta^2\tilde{g}_{AB}\tilde{u}^B = \left(\tilde{u}^C\tilde{\nabla}_C\hat{b}\right)_A = \pm\sqrt{|\Delta|}\beta\left(\tilde{u}^B\tilde{\nabla}_B\tilde{v}\right)_A\,,$$

which is equivalent to the first equation of (B.10). This can be seen by computing the components of either equation and inserting the relationship (B.12).

Now specialising to Schwarzschild, because of the above equivalence, we can find a spherically symmetric solution to the full CGE, parametrised with respect to $\tilde{\tau}$, by solving the first equation of (B.10), given explicitly by computing the Christoffel symbols:

$$\begin{gathered}\tilde{t}'' + \frac{2m}{r^2-2m}\tilde{t}'\tilde{r}' - \frac{\tilde{r}\beta}{\tilde{r}-2m}r' = 0\,,\\ \tilde{r}'' + \frac{m}{\tilde{r}^2}\left(1-\frac{2m}{\tilde{r}}\right)(\tilde{t}')^2 - \frac{2m}{r^2-2mr}(\tilde{r}')^2 - \beta\left(1-\frac{2m}{\tilde{r}}\right)\tilde{t}' = 0\,,\end{gathered}$$

together with the normalisation equation for $\tilde{g}$, (B.5) written explicitly:

$$\frac{\tilde{r}-2m}{\tilde{r}}(\tilde{t}')^2 + \frac{\tilde{r}}{\tilde{r}-2m}(\tilde{r}')^2 = 1\,,$$

where we allow $\tilde{x}'$ to denote differentiation with respect to $\tilde{\tau}$. Combining these equations we can write an equation solely in terms of $\tilde{r}$ as

$$\tilde{r}'' + \frac{m}{\tilde{r}^2} - \beta\sqrt{1 - \frac{2m}{\tilde{r}} + (\tilde{r}')^2} = 0\,, \tag{B.13}$$

which we can integrate with the observation that

$$\frac{\tilde{r}'}{\sqrt{1 - \frac{2m}{\tilde{r}} + (\tilde{r}')^2}}\left(\tilde{r}'' + \frac{m}{\tilde{r}^2} - \beta\sqrt{1 - \frac{2m}{\tilde{r}} + (\tilde{r}')^2}\right) = \frac{\mathrm{d}}{\mathrm{d}\tilde{\tau}}\left(\sqrt{1 - \frac{2m}{\tilde{r}} + (\tilde{r}')^2} - \beta\tilde{r}\right)\,.$$

It follows from the above equality that

$$\tilde{r}' = \pm\sqrt{(C + \beta\tilde{r})^2 - 1 + \frac{2m}{\tilde{r}}}\,, \tag{B.14}$$

for a constant determined by initial conditions[6] $C = \sqrt{1 - \frac{2m}{r_*}} - \beta r_*$. We realise that if there exists a conformal geodesic that is constant in r, it has $r_* = \hat{r}$ that satisfies

$$\frac{m}{\hat{r}^2} - \beta\sqrt{1 - \frac{2m}{\hat{r}}} = 0\,. \tag{B.15}$$

Assuming that this equation has only one solution, $\hat{r}$ the conformal geodesics with $r_* > \hat{r}$ will pass through $\mathscr{I}$ and conformal geodesics with $r_* < \hat{r}$ will move toward the singularity; together with the form of (B.13) this is a hint that these types of congruences of conformal geodesics will be caustic free. Lastly, we derive an expression for the parameter value for conformal geodesics with decreasing r to reach the singularity. As $r_* < \hat{r}$ in this case, we may assume the negative sign holds in (B.14) and this integration is given by

$$\tilde{\tau} = \int_r^{r_*} \frac{\mathrm{d}t}{\sqrt{(C + \beta t)^2 - 1 + \frac{2m}{t}}}\,,$$

from which we can apply (B.8) in order to write

$$\tau = \frac{\underline{\Theta}(2 + \underline{b}_c\underline{u}^c)}{|\beta|}\tanh\left(\frac{|\beta|}{2}\int_r^{r_*} \frac{\mathrm{d}t}{\sqrt{(C + \beta t)^2 - 1 + \frac{2m}{t}}}\right)\,, \tag{B.16}$$

where we have used the identity $x = \tanh\left(\frac{1}{2}\log\left(\frac{1+x}{1-x}\right)\right)$. Then numerical evaluation of the function above for $r = 0$ gives the required value.

[6] Note that $\underline{r} = \tilde{\underline{r}} = r_*$ for some r_* on an initial hypersurface $\underline{S}$.